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A Bias-Variance and Performance Analysis of Imputation Methods for Time Series Forecasting Under Synthetic Missingness Mechanisms

*A study submitted in partial fulfilment of the requirements for the degree of **MSc Data Science***

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Abstract

Background. Missing data is a key issue across many domains, and is widely categorised into three mechanisms of missingness: Missing At Random (MAR), Missing Completely At Random (MCAR) and Missing Not At Random (MNAR). The reconstruction, or imputation, of this missing data risks propagating errors into downstream predictive tasks, such as forecasting for time series data. However, current imputation benchmarking studies focus on reconstruction accuracy, or basic downstream predictive performance, neglecting deep error insight into the predictions.

Aims. This research aims to analyse how each of the missingness mechanisms, along with missingness proportion affects reconstruction performance. This analysis is then extended to examining how systematic error (bias) and instability (variance) are propagated through a range of imputation methods into time series forecasts.

Methods. Using the Electricity Transformer (ETT) dataset, synthetic missingness was repeatedly generated under each of the missingness mechanisms, before being imputed with a range of imputation models, and reconstruction performance aggregated. These reconstructed datasets were then passed to forecasting models, with forecasting errors decomposed into bias and variance.

Results. Reconstruction accuracy declined with increased missingness proportions for all competent imputers, with performance broadly reflective of model complexity; the neural network based method reconstructed most accurately at low missingness but quickly degraded as missingness proportions rose. MCAR was least affected by missingness proportions, while MAR and MNAR suffered from skewed distributions of remaining values. Downstream, the forecaster proved to dictate how errors are propagated into forecasts, with the multivariate forecaster demonstrating much greater bias and variance compared with the univariate, as it was exposed to reconstruction errors across all features. Critically, the strongest reconstructor produced the least stable prediction.

Conclusions. Reconstruction accuracy is not necessarily reflective of forecasting quality. Instead, imputers and forecasters should be regarded as a pair with consideration to the type of errors induced, rather than reconstruction accuracy alone. Future work can extend into synthetic temporal missingness patterns, additional forecasters, and generalise findings to more datasets.

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List of Acronyms

ACF Autocorrelation Function

ADF Augmented Dickey-Fuller

ARIMA Autoregressive Integrated Moving Average

BRITS Bidirectional Recurrent Imputation for Time Series

DL Deep Learning

EDA Exploratory Data Analysis

ES Exponential Smoothing

ETT Electricity Transformer

HUFL High Useful Load

HULL High Useless Load

kNN k Nearest Neighbour

KS Kalman Smoothing

LLM Large Language Model

LOCF Last Observation Carried Forward

LSTM Long Short-Term Memory

LUFL Low Useful Load

LULL Low Useless Load

MAE Mean Absolute Error

MAPE Mean Absolute Percentage Error

MAR Missing At Random

MCAR Missing Completely At Random

MI Multiple Imputation

MICE Multiple Imputation by Chained Equation

ML Machine Learning

MNAR Missing Not At Random

MUFL Middle Useful Load

MULL Middle Useless Load

OT Oil Temperature

PACF Partial Autocorrelation Function

RF Random Forest

RMSE Root Mean Squared Error

RNN Recurrent Neural Network

SARIMA Seasonal Autoregressive Integrated Moving Average

STL Seasonal and Trend decomposition using Loess

SVM Support Vector Machine

TS Time Series

XGB XGBoost

CHAPTER 1

Introduction

This chapter provides a brief summary of the surrounding context for this project (Section 1.1), before framing the aims of this research (Section 1.2.1), the questions it aims to answer (Section 1.2.2), and the objectives required in order to provide answers to these questions (Section 1.2.3).

1.1 Background for the Work

Missing data is an issue across many disciplines (Boros & Kmetty, 2023) that refers to the absence of values within a dataset. Such missingness may arise from sensor malfunction, participant non-response, or data collection errors (Ranganathan & Hunsberger, 2024). Missing data can reduce statistical power, introduce bias, blur sample representativeness and ultimately compromise downstream analyses and model predictions (Kang, 2013; Kelkar, 2022).

Missing data is typically classified into three mechanisms: Missing Completely At Random (MCAR), Missing At Random (MAR), and Missing Not At Random (MNAR) (Rubin, 1976). MCAR occurs when missingness is independent of both observed and missing values. MAR occurs when missingness depends only on observed values, whereas MNAR occurs when missingness depends on the missing values themselves. By understanding the missingness mechanisms, we are able to reduce bias using techniques such as imputation (Groenwold & Dekkers, 2020).

Imputation is the procedure of replacing missing values with plausible substitutes (Adhikari et al., 2022). Imputation methods are commonly divided into single imputation and Multiple Imputation (MI). Single imputation replaces each missing value with a single estimate, whereas MI generates multiple plausible values and combines results across multiple datasets to account for uncertainty (Austin et al., 2021; Dziura et al., 2013). While

more advanced and computationally demanding, MI is a valid approach with many data types (Jakobsen et al., 2017).

The structure of the data impacts the choice of imputation method (Little & Rubin, 2019). Time Series (TS), or temporal, data is particularly nuanced in that imputation with TS data requires interaction with the temporal aspect (Moritz & Bartz-Beielstein, 2017). Since such data is important in many disciplines like signal processing, finance, and climate science, much attention has been paid to developing advanced TS specific imputation techniques (Khan & Lazar, 2023).

Evaluating imputation methods is non-trivial. Performance is commonly measured on reconstruction accuracy, yet imputed data is frequently used in downstream tasks such as statistical inference and predictive modelling (Khademi, 2020). Consequently, different imputation methods may introduce systematic error and instability that propagate into downstream tasks (Engels & Diehr, 2003; Toye et al., 2025), and so a single metric is insufficient.

Despite ongoing developments in imputation techniques and interest in comparative benchmarking of them, studies often focus on naturally occurring missingness and prioritise reconstruction metrics, with limited consideration of deep structural error (Toye et al., 2025). In particular, the bias-variance trade-off (James et al., 2013), which is a central statistical concept, has received relatively limited explicit attention in the evaluation of temporal imputation methods. Given the prevalence of missing data in time series datasets, and the ongoing development in imputation techniques, there is a need for systematic, mechanism-aware evaluation. This project thus adopts a synthetic approach, simulating missing data under different missingness mechanisms and levels, to provide a practically oriented comparison of traditional and modern temporal imputation techniques. It will examine not only reconstruction accuracy, but also downstream forecasting performance through bias and variance analysis.

1.2 Research Aims, Questions and Objectives

1.2.1 Research Aims

The aim of this project is to systematically compare a range of imputation techniques for temporal data under a variety of synthetically controlled missingness mechanisms and levels. Comparison will assess imputation performance in terms of reconstruction, as well as bias and variance analysis of downstream forecasting tasks.

1.2.2 Research Questions

1. How do different missingness mechanisms (MAR, MCAR, MNAR) and missingness proportions affect reconstruction performance of time series imputation methods?
2. To what extent do different time series imputation techniques demonstrate different bias-variance trade-offs within downstream forecasting tasks?

1.2.3 Research Objectives

1. Review and categorise a select number of existing time series imputation techniques, including traditional statistical methods and modern, deep learning methods.
2. Construct controlled experimental datasets using synthetic missingness, emulating MAR, MNAR and MCAR missingness mechanisms across varying missingness proportions.
3. Implement and apply a selection of temporal imputation techniques to the datasets in a consistent experimental framework.
4. Evaluate and analyse reconstruction performance using standard metrics, and analyse downstream forecasting performance, focusing on bias-variance analysis.

CHAPTER 2

Literature Review

This chapter covers a review of the literature surrounding imputation as a whole (Section 2.2.1), TS specific imputation (Section 2.2.2), comparative benchmarking studies (Section 2.2.3), TS forecasting (Section 2.2.4), synthetic missingness generation (Section 2.2.5) and the bias-variance trade-off (Section 2.2.6), before identifying the research gap and position of this research (Section 2.2.7).

2.1 Literature Search Strategy

Literature was identified using strategic searches of Google Scholar and StarPlus. Keywords included ‘imputation’, ‘machine learning’ and ‘missing data’. Well-cited textbooks that are standard references for many topics, cited in many papers, proved to be valuable sources of literature. Furthermore, within certain, more niche topics, using the references from a given paper to explore further was highly effective in identifying important literature within those topics.

All references, aside from those with very high citation counts, were filtered to be peer-reviewed, with a preference towards modern studies where appropriate.

2.2 The Literature Review

2.2.1 Foundations of Missing Data Imputation

Given the challenges that missing data presents, researchers have developed numerous imputation methods with the aim of accurate reconstruction and limiting bias (Emmanuel et al., 2021; Hameed & Ali, 2023; Liew et al., 2011). Historically, imputation focused on single imputation methods until Rubin (1987) developed MI, which was later formalised

into a framework by Little and Rubin (2019). MI techniques are highly adopted and widely accepted as a standard approach in many disciplines (Austin et al., 2021; Jakobsen et al., 2017). Alongside more complicated MI methods, traditional single imputation methods such as regression imputation and hot deck are still employed due to simplicity and low computational cost (Adhikari et al., 2022). These classical and multiple imputation methods have strong statistical support but are generally focused on tabular data as a whole, overlooking the subtleties adjoining TS data.

2.2.2 Time Series Imputation Methods

Imputation with TS data poses unique challenges compared to tabular data given the inherent temporal dependencies between observations (Du et al., 2023; Moritz & Bartz-Beielstein, 2017). The time element means that data is structured, and imputation techniques must account for, and preserve, this sequential nature. Existing approaches to TS imputation range from classical, rule-based approaches through statistical and probabilistic models, and onto modern, machine and deep learning methods.

Classical methods offer simplicity both algorithmically and computationally, with researchers either employing classical methods directly, or combining and embedding the techniques within machine learning approaches (Alejo-Sanchez et al., 2025). These classical methods include mean imputation, linear and spline interpolation, and forward/-backward filling such as Last Observation Carried Forward (LOCF) (Alejo-Sanchez et al., 2025; Pratama et al., 2016; Ribeiro & Castro, 2022). Research into these methods shows that they may reduce the variance within the data and so increase bias, thus potentially limiting their suitability when downstream tasks such as forecasting aim to limit model bias (Engels & Diehr, 2003). In many comparative studies (Ahn et al., 2022; Cao et al., 2018; Li et al., 2024), they are used as a benchmark to contrast with more advanced methods.

Approaches that are more statistically based attempt to exploit the temporal element of the data. State space models and Kalman filtering techniques offer TS specific approaches that are grounded in probability theory by modelling temporal dynamics and recursively updating estimates (Kannegowda et al., 2024; Saputra et al., 2021). Related statistical

methods, such as moving averages, or autoregressive approaches like Autoregressive Integrated Moving Average (ARIMA), further leverage temporal dependencies and provide more robust imputations compared to simpler methods, while remaining more efficient than complex deep learning techniques, particularly on small datasets (Elgohari & Asim, 2025; Hassani et al., 2019).

Machine Learning (ML) methods such as k Nearest Neighbour (kNN), Random Forest (RF), Multiple Imputation by Chained Equation (MICE) and Support Vector Machine (SVM) offer improved flexibility over classical methods, but often fail to exploit temporal dependencies (Li et al., 2024; Pratama et al., 2016; S.-F. Wu et al., 2015). Adaptations to include local time indexes can overcome this limitation (S.-F. Wu et al., 2015). More recently, several substantial, TS specific Deep Learning (DL) architectures have been developed, with many being developed from Recurrent Neural Networks (RNNs) (Fang & Wang, 2020). Methods such as Long Short-Term Memory (LSTM), Bidirectional Recurrent Imputation for Time Series (BRITS) and DeepMVI outperform traditional machine learning techniques (Bansal et al., 2021; Kazijevs & Samad, 2023; Niako et al., 2024). The trade-off comes from computational requirements. ML methods are less demanding but prone to overfitting, while DL approaches are highly computationally intensive and difficult to tune and interpret but offer the strongest performance of all approaches, especially with complex data (Bansal et al., 2021; Jaradat et al., 2024). Moreover, some researchers are experimenting with the use of Large Language Models (LLMs) to impute TS data with remarkable performance (Jacobsen & Tropmann-Frick, 2024; Sartipi et al., 2025; T. Zhou et al., 2023).

Overall, TS imputation methods span a large spectrum, ranging from simple rule-based approaches, to probabilistic and statistical approaches, through to machine learning and highly advanced deep learning techniques. Classical approaches remain widely used due to simplicity and interpretability, but are limited in their ability to incorporate temporal dependencies, and by their introduction of bias. Statistical and probabilistic methods mitigate these limitations to an extent by providing a more principled use of temporal aspects and an overall strong balance between performance and interpretability.

In contrast, machine and deep learning methods emphasise performance, particularly in complex datasets, at the expense of computational cost. Deeper comparisons and benchmarks of these different approaches are required in identifying the most suitable imputation approach, particularly under different missingness mechanisms.

2.2.3 Comparative Benchmarking Studies

Benchmarking imputation methods is a critical component of existing literature surrounding imputation as a whole, with the effectiveness of an imputation method not only being graded on its reconstruction accuracy, but also its impact on downstream predictive tasks (Kazijevs & Samad, 2023). Many studies focus on the former, with common evaluation metrics being Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) (Afrifa-Yamoah et al., 2020; Chakrabarti et al., 2023; Li et al., 2024), with RMSE being the most popular of all (Alabadla et al., 2022). These metrics are easily interpretable and provide a quantitative comparison between methods, forming the basis of many comparative benchmarking studies. It is through these metrics that it is known that MI methods outperform classical methods in terms of reconstruction accuracy (Ahn et al., 2022), but this does not capture other factors that surround imputation performance.

In real world contexts, imputed data is subsequently used in making statistical inferences and predictive models, so discrepancies introduced at the imputation stage can propagate into future decision making. As such, recent research has begun to evaluate predictive performance alongside imputation reconstruction analysis. The TSI-Bench framework by Du et al. (2024) explored regression, classification and forecasting predictive tasks using imputed data, concluding that the choice of imputer measurably impacts downstream forecasting performance. TS specific forecasting in particular has further been incorporated into benchmarking studies (Du et al., 2024; Elgohari & Asim, 2025; Poette et al., 2026).

Despite this development, the connection between reconstruction accuracy and downstream effectiveness remains underexplored. Even very recently, in a thorough benchmark

of TS imputation methods under varying missingness mechanisms and proportions, Toye et al. (2025) explicitly highlight the need for further analysis on the downstream impact of imputation.

2.2.4 Time Series Forecasting Methods

TS forecasting refers to the act of predictive modelling using TS data, relying on past observations to predict the future (Hyndman & Athanasopoulos, 2021). The length of this forecast is referred to as the horizon. Forecasting methods, like imputation methods, range across statistical and ML approaches (Makridakis et al., 2018).

Hyndman and Athanasopoulos (2021) describe a variety of statistical forecasting methods, many of which aim to exploit recurring patterns, referred to as seasonality, and long term behaviour, called trends, within the TS data. Many of these methods, including ARIMA and Exponential Smoothing (ES), are univariate and linear in form, modelling the target feature purely by its own history. Holt-Winters, an extension of ES, for example, models a series through a combination of separate level, seasonal and trend components (Hyndman et al., 2002). It is crucial to identify such behaviours of a TS when setting the parameters of these models; techniques such as Seasonal and Trend decomposition using Loess (STL) allow for the TS to be deconstructed into separate components that then inform model specification (Krupakar & Kandappan, 2025). Such methods are valued for their interpretability and low computational demand, performing particularly well on highly seasonal data with short forecast horizons (Hyndman et al., 2002).

ML methods, conversely, do not explicitly exploit seasonality or trends. Instead, forecasting is reframed as a regression problem, with future values being predicted from engineered calendar and lag features derived from past observations (Hyndman & Athanasopoulos, 2021). This inherently allows ML methods to be multivariate but does necessitate feature engineering as standard. Tree-based gradient boosting, namely XGBoost (XGB) (Chen & Guestrin, 2016), is a well-founded choice for tabular forecasting, with strong performance highlighted in forecasting competitions (Januschowski et al., 2022). In more recent times, DL methods including RNN, LSTM, GRU and Transformer models

have grown rapidly in importance within the domain of TS forecasting, handling modern data volume demands and model complexities with strong predictive capabilities, at the cost of computational requirement (Song et al., 2024; Yu et al., 2025). Evidence suggests, however, that through window-based input transformations, gradient boosting models can outperform deep neural networks in TS forecasting tasks (Elsayed et al., 2021).

The relative performance of ML and statistical methods is not fixed, and instead is highly dependent on the data and its characteristics. Using a large subset of monthly time series M3 competition data, Makridakis et al. (2018) observed superior forecasting accuracy from statistical models over ML methods, with greatly reduced computational cost. The M5 competition saw a reversal, where ML methods consistently outperformed statistical methods (Makridakis et al., 2022). This variability in performance between the classes of forecasting methods motivates the need for representatives of both within experimentation.

For any forecast where the horizon is greater than a single step, a multi-step strategy must be employed. A common strategy is to iteratively create forecasts one step at a time, by feeding predictions back into the model to then predict the next step (Ben Taieb & Atiya, 2016). Ben Taieb and Atiya (2016) demonstrate, however, that these methods exhibit structural errors by inputting forecasting errors into the model, accruing bias. Direct approaches, conversely, where forecasting steps are predicted independently, suffer from higher variance. The notion of this trade-off between bias and variance is expanded upon further in Section 2.2.6.

2.2.5 Synthetic Missingness

Evaluating the reconstruction accuracy of an imputation method requires a comparison between the imputed and true values, thus requiring knowledge of the data that is missing. This is clearly a dilemma when using naturally occurring missing data and thus researchers simulate missingness within complete datasets, a process referred to as amputation (Schouten et al., 2018). The amputation step defines the conditions under which imputation methods are assessed, hence a poorly defined procedure risks invalidating an experiment's

conclusions (Santos et al., 2019).

Existing amputation procedures are broadly categorised in two ways. Stepwise univariate amputation, as named by Schouten et al. (2018), generates missingness one feature at a time, generating the missingness in one variable independently and separately of any others. The alternative is pattern-based generation. A missingness pattern specifies a particular combination of variables in which to introduce missingness, with multiple patterns defined across a complete dataset before amputation (Schouten et al., 2018). Santos et al. (2019) further divide the stepwise approach according to where the resulting missingness lies. Univariate configurations contain all the missingness within a single feature, with multivariate configurations distributing the missingness across several features. Univariate here refers to the outcome of the amputation, contrasting to the stepwise univariate approach that applies to one variable at a time, leading to multivariate amputation.

Regardless of the configuration, each missingness mechanism must be simulated differently; Santos et al. (2019) summarise many existing approaches. MCAR is typically generated through a uniformly random deletion of values, such as using a random number generator to identify locations of values to be deleted, or by repeated Bernoulli trials on each feature, with the probability of missingness defined by the desired missingness proportion. MAR implementations typically rely on a causative feature, from which observed values dictate the missingness of values in another. The mapping from the driver feature to missingness is achieved in various ways, including percentile cut-offs, rankings, and logistic functions. MNAR implementations condition missingness on the values themselves and are largely deterministic, deleting extreme values against some threshold to meet the desired missingness proportion.

A persistent and crucial issue across these approaches is the failure to accurately generate missingness proportions. Schouten et al. (2018) demonstrated that when features are amputated independently, in stepwise univariate amputation, the resulting joint proportion of missingness is not the desired one; simulating 50% MCAR missingness in two different features results in only 25% missingness in both, while the equivalent MAR approach using a standard logistic function results in 29% joint missingness. A MNAR

experiment intended to produce the same 50% missingness produced missingness in only 19% of cases. Shifting the logistic function can be a correction for the discrepancy but requires extensive, complicated mathematics and importantly alters the behaviour of the mechanisms, skewing observed distributions.

Santos et al. (2019) identified further theoretical and empirical flaws, alongside potential experimental hazards associated with current methods. Large sample sizes are required in order for Bernoulli trials to converge to the correct missingness proportion, leaving smaller datasets prone to over- or undershooting the desired missingness. With percentile cut-off approaches, features that contain repeated values may fail to remove the correct proportion as the repeats impact the feature's distribution. Proportions may also be limited to 50 or even 25% when the missingness is restricted to a subset of features. An important observation from both reviews is the potential collapse of a mechanism entirely. Where there are weak relationships between a driver and its target, or when using median groupings, MAR mechanisms can effectively behave as MCAR, although the latter situation is more prominent in datasets that contain qualitative features.

Researches have developed packages with the aim of consolidating current amputation procedures. Schouten et al. (2018) implement multivariate amputation within the mice R package, in which the user specifies the missing data patterns and feature weights, with weights determining the mechanism. Schouten et al. (2022) create a Python library of many amputation methods, pyampute, while Mangussi et al. (2025) created mdatagen, a library that implements the uni- and multivariate strategies as catalogued by Santos et al. (2019).

Despite the consolidation, existing approaches commonly perform amputation on a feature-wise basis, rather than at an individual cell level, resulting in issues in achieving accurate missingness proportions. Furthermore, these amputation procedures are targeted towards tabular data and thus do not incorporate temporal aspects. This leaves time series specific behaviours, such as consecutive missing values, unrepresented.

2.2.6 Bias-Variance Trade-Off

The bias-variance trade-off is a central concept in machine learning that decomposes the error of a model's prediction into distinct sources (James et al., 2013). For a trained model, on a pointwise basis, the expected squared error separates into the three components: the squared bias, the variance, and an irreducible error. Bias is the systematic error introduced by a model misrepresenting the underlying relationships in the data, and instead remaining inflexible. High bias is a common characteristic of simple models that underfit the data. Contrastingly, variance describes a model's sensitivity to its training data. A model demonstrating volatility in the predictions between different training samples is described as overfitting, producing high variance. High variance is more likely to be seen in models that are flexible, with models capturing sample-specific relationships rather than generalised patterns. The irreducible error is simply noise introduced at the data generating stage that no model can reduce.

The two reducible components, bias and variance, cannot generally be minimised simultaneously. As a model's flexibility increases, so does its variance but its bias decreases. Conversely, as a model's flexibility decreases, its bias is increased but its variance decreased. This characteristic trade-off leads to a desired optimal balance of model flexibility that minimises error (James et al., 2013). Decomposing errors in such a way offers greater insight beyond a single performance metric, allowing users to understand the source and nature of errors, whether this is due to systematic rigidity or instability with different data, hence ascertaining whether or not a model is under- or overfitting (Belkin et al., 2019; Guan & Burton, 2022; Rosset & Tibshirani, 2020).

Despite its centrality to the evaluation of predictive models, the bias-variance trade-off has received comparatively little attention within imputation research, where, as discussed in Section 2.2.3, imputation evaluation is largely dominated by reconstruction metrics. Within TS forecasting, Ben Taieb and Atiya (2016) applied a bias-variance decomposition to multi-step forecasts, where errors in the recursive strategies were attributed to bias compared with direct strategies demonstrating higher variance. While this grounds the

need for bias-variance decomposition within forecasting, the extension of this to imputation remains largely unexplored. Different imputation methods can be seen as different sources of variability, with different reconstructions of the same data potentially propagating systematic errors into downstream forecast tasks. It is the current gap in analysis of this structural error propagation of imputation methods into forecasting that this work addresses.

2.2.7 Gap and Position

The preceding sections reveal several gaps within existing research. Much imputation research is domain specific, and thus evaluations take place against real-world missingness. While this grounds research in realistic scenarios, it does not preserve the controlled, systematic nature of experiments conducted with synthetic missingness. Where synthetic missingness is employed, however, studies primarily focus on the MCAR mechanism (Y. Zhou et al., 2024), leaving the more complex mechanisms MAR and MNAR as comparatively under-represented within controlled experiments. Furthermore, as established in Section 2.2.5, current amputation procedures operate on feature-wise or pattern-wise bases rather than cell-wise, and struggle to accurately preserve missingness proportions.

A further gap regards how imputation is evaluated. Reconstruction metrics such as RMSE and MAE dominate the literature (Section 2.2.3) and offer quantitative and comparable evaluation, but are restricted to only assessing the recovered values in isolation. Imputation, in practice, aims to recreate missing data so that it may then be used in downstream tasks such as for statistical inference, including forecasting. While recent benchmarks have begun to incorporate forecasting, the general relationship between reconstruction accuracy and downstream predictive ability remains insufficiently examined. It remains to be seen if the most accurate reconstruction necessarily leads to optimal forecasts.

Additionally, the structural characteristics of downstream forecasting error have received little attention. Reconstruction metrics offer no insight into how an imputation method may introduce systematic bias or instability into subsequent tasks. In particular, the bias-variance trade-off, despite being a foundational tool for understanding predictive

errors, has scarcely been applied to imputation, notably within the time series domain where forecasting is the primary downstream task of interest.

The research presented here aims to address these gaps through a systematic benchmarking framework. Traditional and modern time series imputation methods are methodically compared across all three missingness mechanisms, generated synthetically for experimental control, under a range of missingness proportions. Evaluation will not only be conducted on reconstruction accuracy, but additionally on the subsequent forecasts through a bias-variance analysis.

CHAPTER 3

Methodology and Implementation

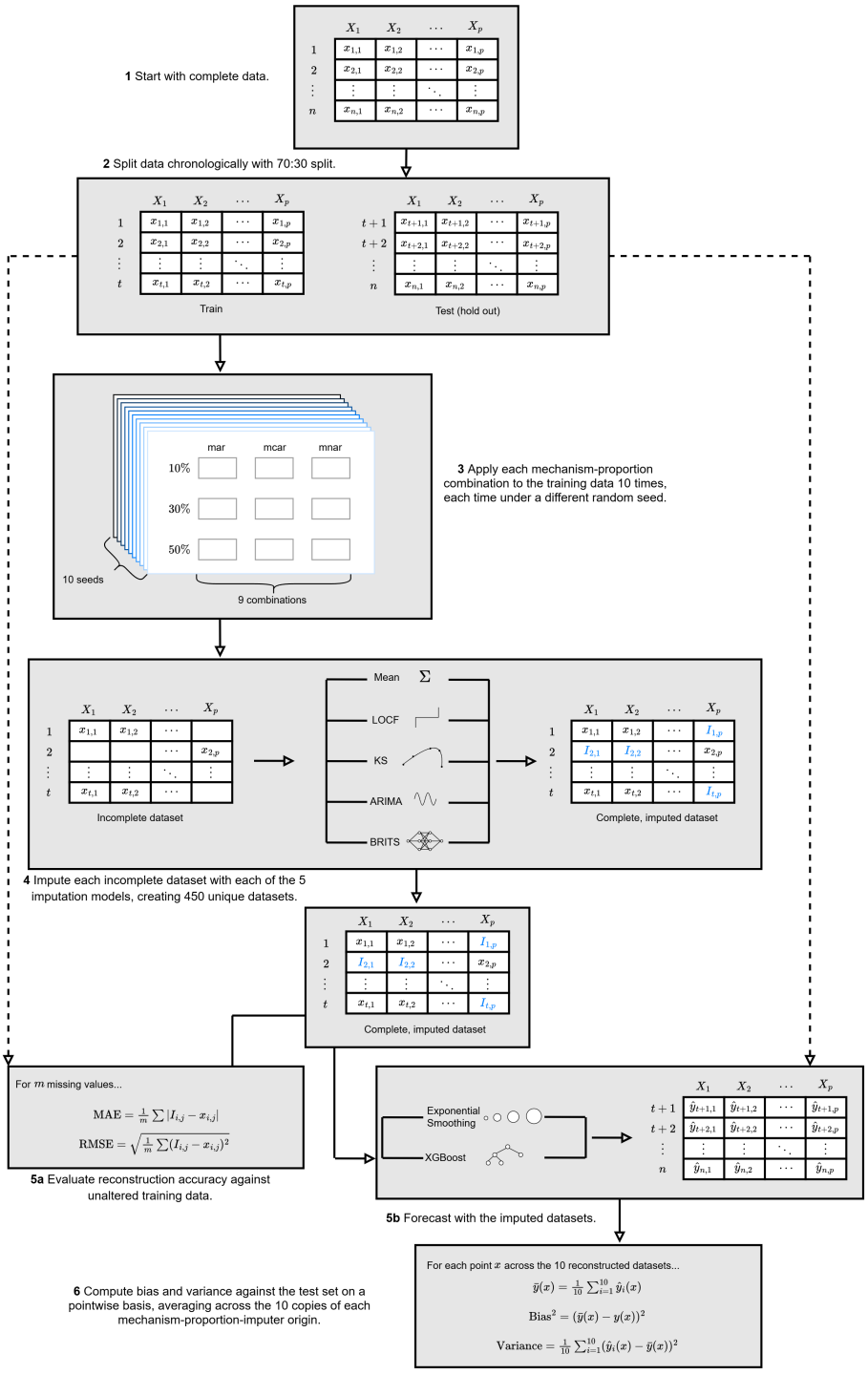


Figure 3.1. Data pipeline and overview of methodological approach.

Figure 3.1 depicts the overall methodological steps taken in this experiment. The original, complete dataset was split chronologically using a 70% training proportion, with both the complete training and complete test data preserved for evaluations further down the pipeline.

At step 3 in the diagram, the training set was amputated. This amputation was conducted under each missingness mechanism MAR, MCAR and MNAR, each at a proportion of 10%, 30% and 50% missingness. As discussed further in Section 3.1.1, each of these mechanisms is stochastic. This allowed amputation at each mechanism-proportion combination to be conducted under 10 unique random seeds, producing a unique amputation pattern, necessary for later bias and variance computations. This amputation was applied using masks, where $3 \times 3 \times 10 = 90$ unique masks were created, for each of the 3 mechanisms and proportions, and 10 seeds.

The 90 unique, incomplete training datasets created in step 3 were then passed to 5 imputation models: mean, LOCF, Kalman Smoothing (KS), ARIMA and BRITS. The result of passing each of the 90 incomplete datasets to each of these models was $90 \times 5 = 450$ unique, reconstructed datasets. The reconstruction accuracy of these datasets can be measured against the original, complete training data. By masking the original data, MAE and RMSE can be calculated only using the reconstructed cells and their true values. This evaluation will answer Research Question 1 (Section 1.2.2).

Additionally, these 450 reconstructed datasets were used to train 2 forecasting models: ES and XGB. This produced $2 \times 450 = 900$ forecasts, originating from the mechanism, proportion, seed and forecaster combination. As there are 10 versions of each mechanism, proportion and forecaster combination, each of a different random seed, variance is calculated as an average spread of the predictions across these 10 datasets. Mean bias requires collapsing these 10 datasets and measuring against the original test data, as well as forecasts produced by the forecaster models trained on the original, complete training data, directly answering Research Question 2 (Section 1.2.2).

3.1 Methodological Approach

3.1.1 Synthetic Missingness

Missing data in the dataset was simulated under the three mechanisms of missing data: MAR, MCAR and MNAR (Rubin, 1976). In this section, the mathematical properties of each type of missingness will be explored, alongside some real world examples to provide context.

As discussed in Section 2.2.5, many approaches result in incorrect proportions of missingness, and the missingness is conducted on a feature-wise basis. The methods of synthetic missingness adopted in this research address these issues by using a cell-wise approach, while still taking inspiration from existing methods of simulating the different mechanisms. In particular, each simulated mechanism selects $N = \lfloor \alpha/100 \cdot n \cdot p \rfloor$ cells to be made missing, where α is the target missingness percentage, and n and p are the number of rows and columns in the data, respectively. This allows for consistent and accurate missingness proportions across all proportions and mechanisms, with the probabilities being determined by the missingness mechanism.

Before proceeding, it should be noted that these approaches of synthetic missingness do not incorporate the temporal aspects of the data. Block missingness is an example of missingness specific to time series, where sequential observations are not present. Through their random nature, however, blocks of missingness do occur, particularly with the higher missingness proportions.

Aligned with Little and Rubin (2019), to formalise the mechanisms, we define a dataset X that can be split into its observed, X_{obs}, and its missing, X_{miss}, values. We further define ψ to be some unknown parameters of the mechanisms.

3.1.1.1 Missingness Masks

A missingness mask M for a dataset X with n rows and p columns, is an $n \times p$ matrix with the following property:

$$M = \{m\}_{i,j=1}^{n,p} = \begin{cases} m_{i,j} = 0, \text{ if } x_{i,j} \text{ is observed,} \\ m_{i,j} = 1, \text{ if } x_{i,j} \text{ is missing.} \end{cases} \tag{3.1}$$

For this experimental design, each missingness mask is constructed for a mechanism-proportion pairing, with a unique random seed for each of the 10 repeats. Using masks in this way greatly reduces computational demand as only 90 unique masks require generating.

3.1.1.2 Missing At Random

The MAR mechanism has a misleading name, in that it is not random but instead, the missingness is dependent on the observed data, as in Equation 3.2.

$$P(M|X, \psi) = P(M|X_{obs}, \psi) \tag{3.2}$$

A simple example of this would be if one sex were less likely to report their weight in a survey where both sex and weight were recorded, if the value of the weight itself had no effect on this likelihood.

Figure 3.2 outlines the procedure for creating the MAR masks. Adapted from approaches proposed by Twala (2009), Pan et al. (2015), Ali and Omer (2017) and Zhu et al. (2012), this method relies on driving missingness within a feature through its most correlated other feature. It is necessary to use the most correlated feature to drive missingness, as low correlation between paired off features tends towards an MCAR mechanism (Oberman & Vink, 2024).

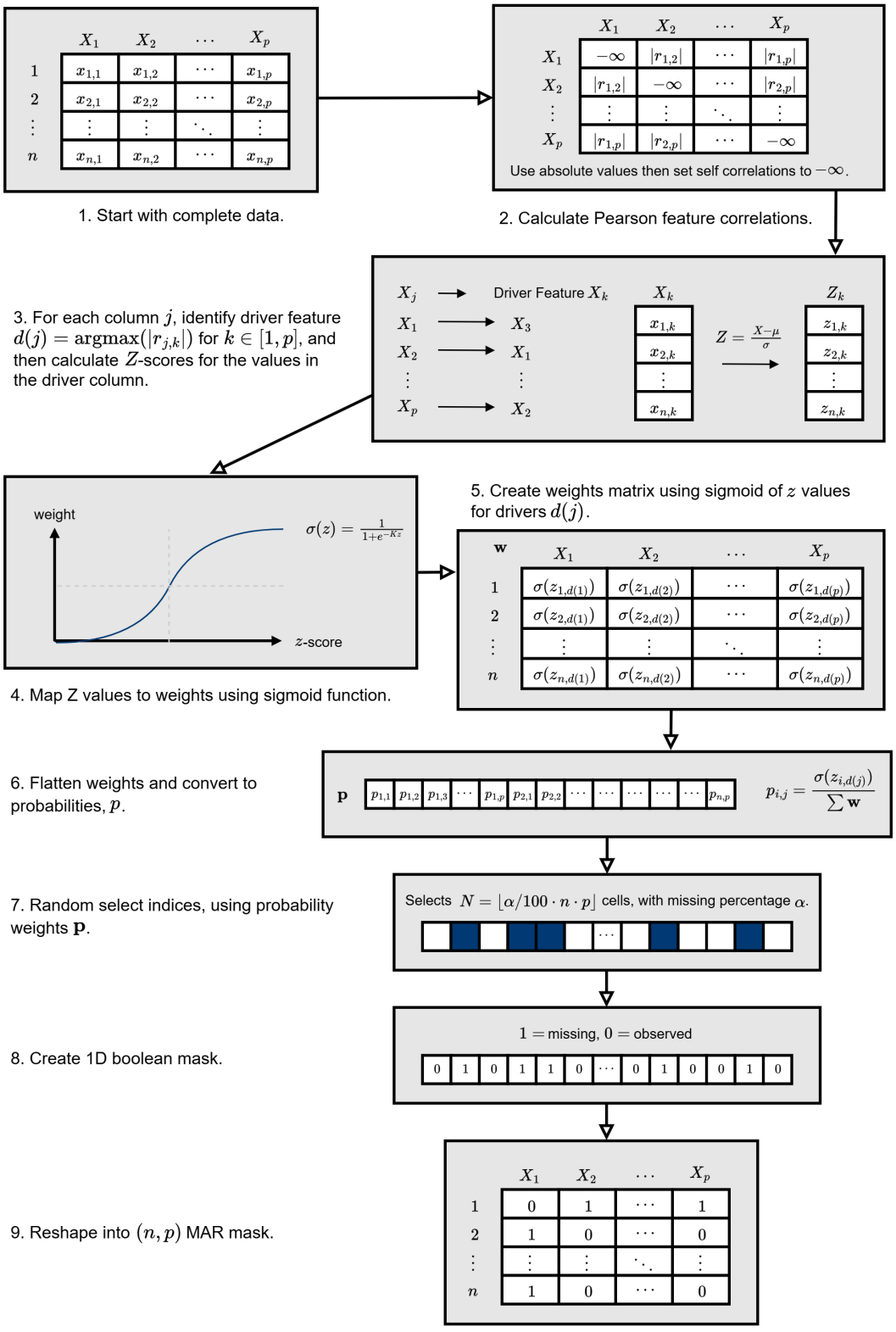


Figure 3.2. Diagrammatic representation of simulating MAR missingness: corresponding code in Appendix E.1.

The Z-scores of each value in the column containing the missingness can be thought

of as that value's desire to be made missing, converted into weights using the sigmoid function. The sigmoid function implementation here contains a variable K, the logistic steepness, which dictates how strongly the mechanism concentrates missingness. For this experiment, $K = 4$. Converting these weights into probabilities creates a probability distribution within the data, for the probability of missingness. This distribution is used for the random index sampling to then create the mask.

3.1.1.3 Missing Completely At Random

Arguably the most simple of the three, MCAR is the mechanism that is truly random. As represented by Equation 3.3, the probability of a value being missing is entirely independent of the data containing the missingness, and only dependent on the unknown parameters ψ.

$$P(M|X, \psi) = P(M|\psi) \tag{3.3}$$

An example of this would be missing values in a study for values recorded at the start of the day, due to participant lateness. This randomness allows for the simulation of this missingness to be a uniformly random selection of cells within the data. By flattening the data into a 1-dimensional array, this selection is multivariate, as shown in Figure 3.3. While not required here, were a weights matrix to be present, every value would be equal to adhere to the uniform distribution. Furthermore, this implementation of MCAR simulation is on a dataset basis, rather than column-wise. This approach therefore reduces the rounding errors in calculating overall missingness proportions compared to other approaches, such as proposed by Twala (2009), that focus on equal missingness proportions across all features.

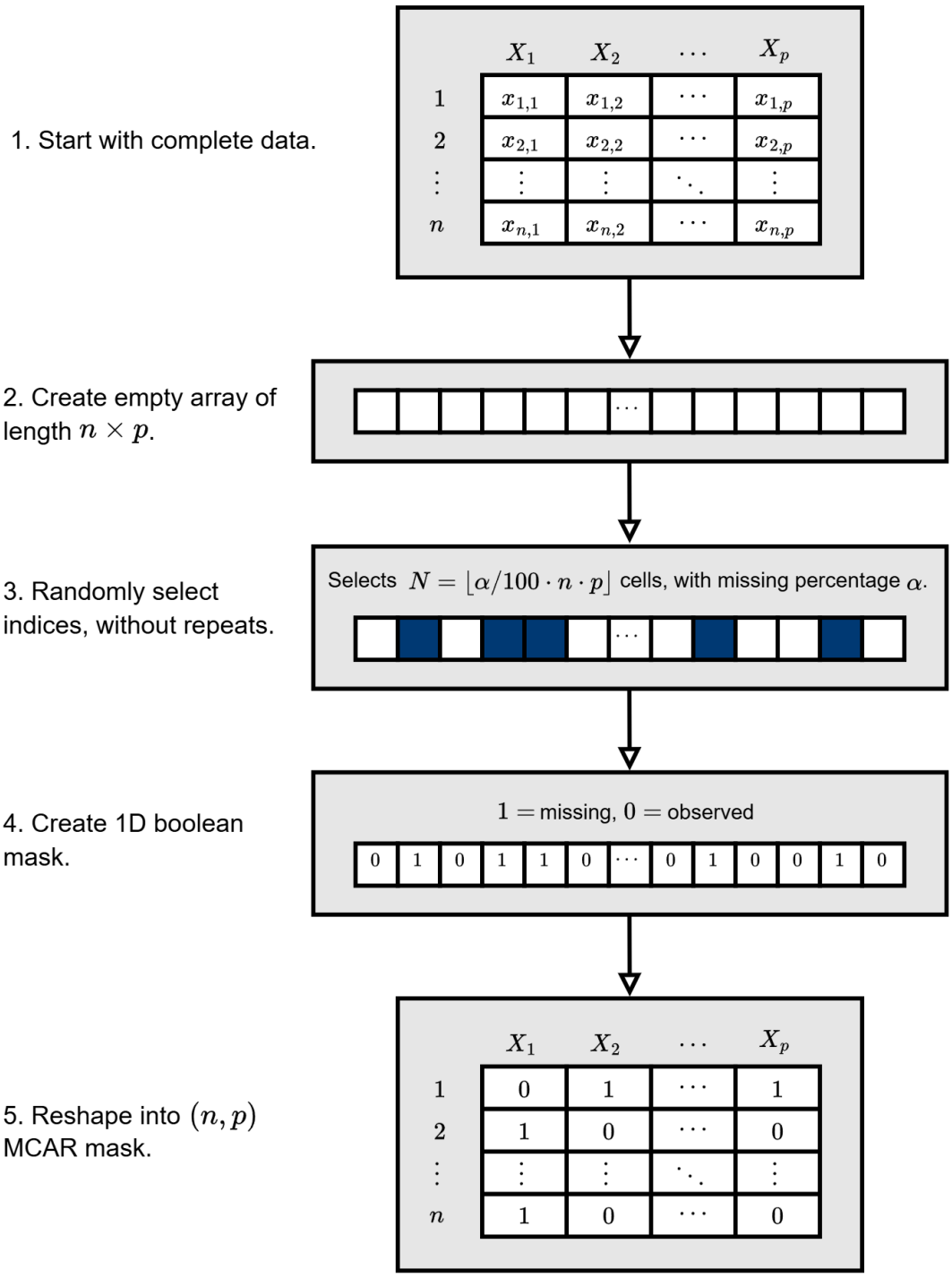


Figure 3.3. Diagrammatic representation of simulating MCAR missingness: corresponding code in Appendix E.2.

3.1.1.4 Missing Not At Random

The final mechanism, MNAR, is described as where the missingness is dependent on the value of the missing data itself, or on other observed values, or on the unknown parameters, such as sensors failing above a certain value.

$$P(M|X, \psi) = P(M|X_{obs}, X_{miss}, \psi) \tag{3.4}$$

Given the complexity of not at random missingness, the approach implemented here restricts the mechanism to focus on generating missingness based on the missing value, depicted in Figure 3.4. This is a stochastic approach compared to the largely deterministic approaches discussed in Section 2.2.5, and closely reflects the MIV (Missingness depending on its Value Itself) method proposed by Garciarena and Santana (2017).

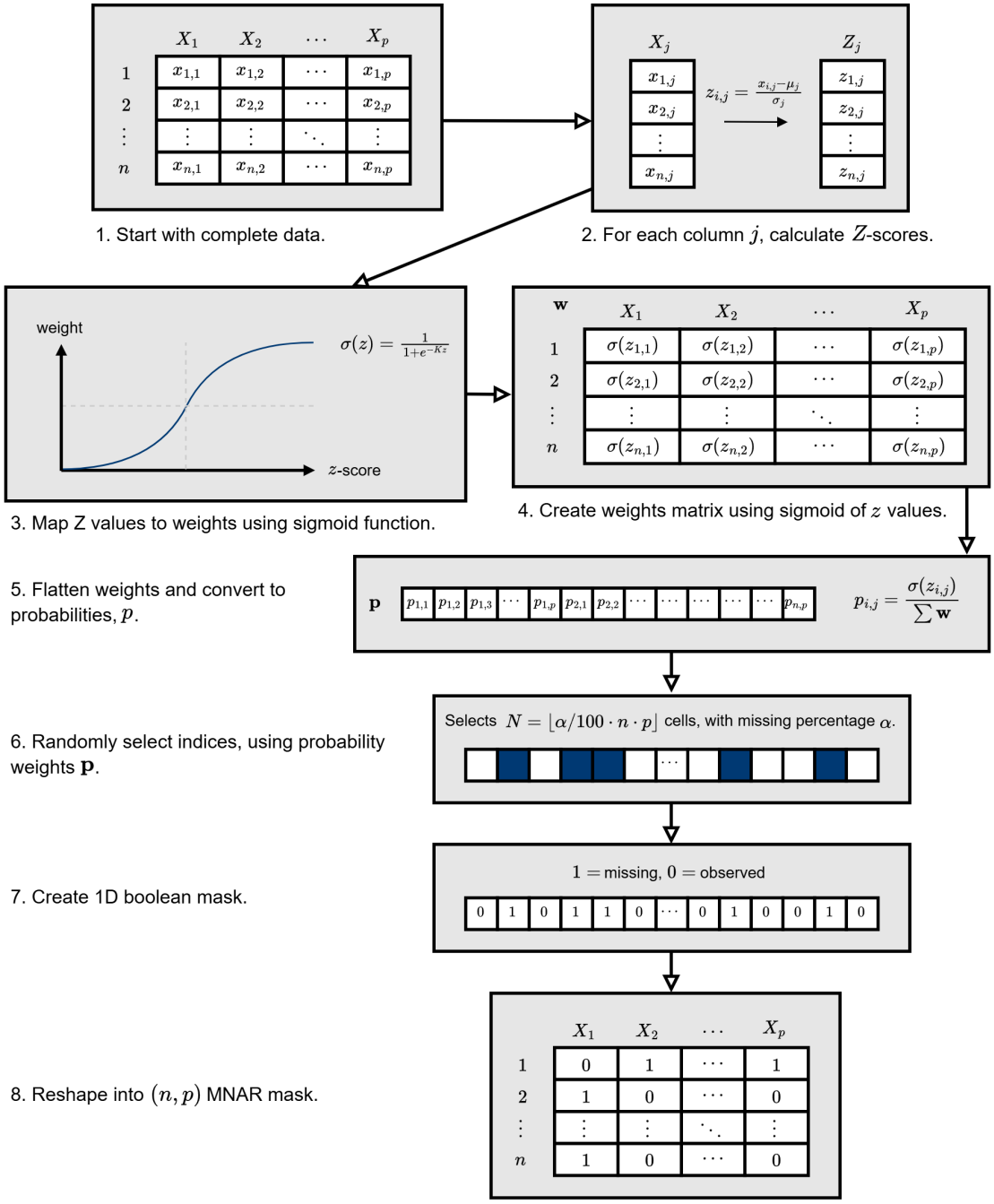


Figure 3.4. Diagrammatic representation of simulating MNAR missingness: corresponding code in Appendix E.3.

3.2 The Data

For this research, a TS dataset is required for imputation and subsequent forecasting.

As this research adopts a synthetic missingness approach, it is important for the dataset

to not contain any missing values. The Electricity Transformer (ETT) dataset meets this criterion (H. Zhou et al., 2021).

3.2.1 Data Overview

First introduced by H. Zhou et al. (2021), the ETT datasets are commonplace in time series benchmarking and deep learning research (Woo et al., 2022; H. Wu et al., 2021; T. Zhou et al., 2022); this research in particular uses the ETTh1 dataset. This dataset is publicly available data recorded over a two year period on an electricity transformer, including electricity loads and oil temperature. The data is measured at hourly intervals between July 2016 and July 2018.

The Oil Temperature (OT) of transformers is reflective of their condition, and predicting the oil's temperature supports load management and prevents damage to transformers (H. Zhou et al., 2021). OT is the target variable within the forecasts, with the remaining features shown in Table 3.1. Each of these features is continuous, and thus only numeric imputation and forecasting methods are required.

Table 3.1. Description of each variable in the ETT dataset.

Variable	Description	Type
date	The recorded date	Date
HUFL	High Useful Load	Continuous
HULL	High Useless Load	Continuous
LUFL	Low Useful Load	Continuous
LULL	Low Useless Load	Continuous
MUFL	Middle Useful Load	Continuous
MULL	Middle Useless Load	Continuous
OT	Oil Temperature (target)	Continuous

The data contains 17,420 observations and, crucially, does not contain any missing values. Additionally, this dataset contains real sensor data, within which are genuine seasonal patterns, trends and irregularities, embedding this research in real world applicability.

3.2.2 Data Preprocessing

Following the chronological split of the data into its training and testing sets, as depicted in Figure 3.1, the data was min-max scaled. Reconstruction metrics are evaluated on a multivariate basis, hence if features were to be of different scales, then errors in different features would carry different weights. Thus, by scaling all values between 0 and 1, aggregations across all features allow for comparable results across the 90 unique missingness patterns and 450 reconstructed datasets. It should be noted, however, that the predicted values of OT from the forecasts were unscaled in order for the temperature values to be comparable with the unaltered test data. To prevent data leakage, the scaling was only calculated from the values in the training set.

3.3 Analysis Methods

3.3.1 Imputation Methods

The imputation methods chosen for this research are a representative sample of simple, statistical and DL methods. As discussed within each of their subsections, the implementation of these methods in Python was relatively straightforward with the use of external packages. There are numerous other imputation methods that could be used here, but each additional method used creates 90 new reconstructed datasets. Hence, due to computational limitations, the methods are restricted to the following.

3.3.1.1 Mean

Mean imputation is commonly used as a naïve baseline in imputation benchmarking research (Poette et al., 2026; Toye et al., 2025). The implementation of mean imputation here was on a column wise basis, where, for each feature, any missing values were imputed with the mean of the observed values.

3.3.1.2 Last Observation Carried Forward

LOCF, also known as forward filling, is a TS specific simple method that imputes a missing value with the previous observed value in that feature. More formally, if the value y_t is missing, then $y_t = y_{t-1}$. If a consecutive string of values is missing, then $y_t = y_{t-n}$ where y_{t-n} is the most recently observed value.

3.3.1.3 Kalman Smoothing

KS treats the observed time series as a noisy observation of a latent state that evolves according to a state space model (Durbin & Koopman, 2012). This model is in essence a pair of equations. The state equation describes the transition of the latent state from one time step to the next, while the observation equation relates that state to the observed series.

A forward pass, the Kalman filter, recursively predicts and updates estimates, before a backwards pass, the Kalman smoother, revises the state estimates using future observations. This allows each point to be informed by the whole series rather than just previous observations.

Imputation follows naturally from these passes. At a missing value, there is no observation to correct against, and so the update step is skipped and the value is generated from the model dynamics and surrounding data, in both directions.

The model is implemented using Unobserved Components from the statsmodels package (Seabold & Perktold, 2010). Treating each column as an individual TS, the local level model is implemented in its simplest structural form, using neither trend nor seasonal components. This provides a statistically grounded and comparatively light imputation approach. The local level form, however, does not capture seasonal or trend structure, hence is unable to reconstruct patterns such as daily cycles where longer missingness gaps are present.

3.3.1.4 Autoregressive Integrated Moving Average

ARIMA is a statistical model that consists of three mechanisms: autoregression, integration, and a moving average (Box et al., 2016). The autoregressive component refers to the model using regression on its own recent past, and the moving average component uses errors of past forecasts in a regression-like model (Hyndman & Athanasopoulos, 2021). The autoregressive and moving average components necessitate that the TS is stationary: the statistical properties of the TS are independent of the time of observation (Hyndman & Athanasopoulos, 2021), and it is the integrated component of the model that removes trend and seasonality to induce this stationarity. The parameters of the ARIMA model p, d and q are the order of the autoregressive component, the order of differencing, and the order of the moving average component, respectively.

Despite these components, ARIMA is unable to incorporate seasonality; Seasonal Autoregressive Integrated Moving Average (SARIMA) is an extension of ARIMA that resolves this shortcoming. By adding a parallel set of seasonal terms P, D and Q, which are seasonal counterparts to p, d and q, SARIMA is able to model repeating patterns, such as daily cycles, for some seasonal period m. Thus the full SARIMA model can be written as

$$\text{SARIMA}(p, d, q)(P, D, Q)_m, \tag{3.5}$$

with these parameters driven by the data (Box et al., 2016; Hyndman & Athanasopoulos, 2021). Imputation using SARIMA mirrors that of KS. The fitted model produces estimates at the missing positions using model dynamics and surrounding values.

This model remains univariate, fitting models on a column-wise basis. Model implementation was through SARIMAX in the statsmodels package (Seabold & Perktold, 2010), with fixed parameters based on exploratory analysis (Section 4.1). As SARIMA is an extension of ARIMA, the results presented in this research will reference the ARIMA model, but note that it is the seasonal extension being used.

3.3.1.5 Bidirectional Recurrent Imputation for Time Series

Developed by Cao et al. (2018), BRITS is a DL imputer based on RNNs. Unlike the statistical methods used in this research, BRITS makes no assumptions about the data generating process. Instead, BRITS learns the missing values directly from the observed data within a bidirectional recurrent dynamical system. This allows the method to handle non-linear underlying dynamics, as well as being multivariate: a trait not possessed by any other imputer used here.

Within the algorithm, missing values are treated as variables of the RNN graph, estimated from the hidden state and then refined through backpropagation. By running both forwards and backwards passes, the bidirectional nature of BRITS mitigates issues of bias exploding and slow convergence in training, enforcing consistency between the directions. BRITS is multivariate, using a feature's own history as well as exploiting correlations across features at the same timestamp.

BRITS was implemented through the Python package PyPOTS (Du et al., 2025). The imputer further required the data to be windowed into fixed length multivariate sequences, with the window length being determined by the seasonal patterns found in Exploratory Data Analysis (EDA) (Section 4.1). As is customary with neural network based models, BRITS is highly computationally demanding and requires sufficient observed data to train on, making it susceptible to high missingness proportions.

3.3.2 Reconstruction Analysis

Reconstruction analysis of imputation methods was calculated by comparing imputed values to their original counterpart, from the original training set. This was achieved by masking the original dataset with the same missingness mask used to ampute the then reconstructed dataset. The standard metrics MAE and RMSE were used to measure imputation performance. For m imputed values,

$$\text{MAE} = \frac{1}{m} \sum_{(i,j) \in M} |I_{i,j} - x_{i,j}| \tag{3.6}$$

$$\text{RMSE} = \sqrt{\frac{1}{m} \sum_{(i,j) \in M} (I_{i,j} - x_{i,j})^2}. \tag{3.7}$$

These metrics offer slightly different perspectives, in that MAE simply measures the average distance from imputed values to their true ones, whereas RMSE adds a penalty for values that are further away. As the amputation is across multiple features, herein lies the need for the scaling discussed in Section 3.2.2, allowing these aggregate metrics.

3.3.3 Forecasting Methods

The forecasting methods are used to predict OT for a 1 week horizon, that is, 168 observations.

3.3.3.1 Exponential Smoothing

ES is a univariate forecasting model that produces forecasts using weighted averages of past observations, with weights further in the past decaying exponentially, assigning more importance to recent observations (Hyndman & Athanasopoulos, 2021).

The Holt-Winters (Holt, 2004; Winters, 1960) extension of the ES model comprises a forecast equation, alongside smoothing equations for level, trend and seasonality. The trend and seasonal components are either additive or multiplicative, the choice of which is dependent on whether seasonal variations are constant, or proportional to the level of the series (Hyndman & Athanasopoulos, 2021). Furthermore, it is possible to add damping to the model, given that longer horizon forecasts tend to over-forecast (Gardner & McKenzie, 1985).

The Exponential Smoothing model from statsmodels (Seabold & Perktold, 2010) was used to forecast OT. The models were fit to the reconstructed OT columns alone, as ES is univariate and only forecasts using the target variable's own history, and no other features.

3.3.3.2 XGBoost

XGB is a gradient boosted tree ensemble (Chen & Guestrin, 2016), where trees are fit sequentially and correct errors from previous ones. XGB is a multivariate algorithm designed to handle tabular data, hence unlike ES, it does not natively handle temporal components like trend and seasonality. Instead, forecasting with tree based models requires reframing the forecast as a regression problem, and engineering features accordingly (Januschowski et al., 2022).

Here, the XGBRegressor was used with calendar features (hour, day of the week and month), alongside lagged values for each of the remaining features. This multivariate prediction exposes XGB to imputed values across all features. Any features at the same timestamp as the intended forecast were removed to prevent data leakage.

As opposed to forecasting every feature individually and then building predictions for OT from these predictions, which would be highly computationally demanding across the 450 datasets, a recursive strategy is used. Forecasts are produced one step at a time, with predictions fed back into the model to predict the following step. As discussed in Section 2.2.4, this allows early errors to propagate into later forecasts.

As OT is the only variable forecast recursively, while the others are held at their last observed values, the lagged features derived from those static columns become constant once the length of the forecast exceeds that of the lag. Naturally, shorter lags will freeze first, but any frozen features compromise the quality of long term forecasts by removing temporal variation.

3.3.4 Forecasting Analysis - Bias-Variance Trade-off

Recall from Section 2.2.6 that prediction errors can be decomposed into squared bias, variance and irreducible error (James et al., 2013):

$$\text{Error} = \text{bias}^2 + \text{variance} + \text{irreducible error}, \tag{3.8}$$

and that this equation describes how predictive models behave in terms of over- and underfitting.

The irreducible error term is not estimated as it represents inherent randomness within the data generating process, as opposed to errors induced in the experimental pipeline. As 10 datasets are reconstructed for each mechanism, proportion, imputer combination, the variance naturally arises as the variance over these 10 datasets in terms of forecasting predictions. Formally, across the datasets, on a pointwise basis, the mean prediction is

$$\bar{y}(x) = \frac{1}{10} \sum_{i=1}^{10} \hat{y}_i(x), \tag{3.9}$$

for each point x. It follows that, again on a pointwise basis,

$$\text{Var}(x) = \frac{1}{10} \sum_{i=1}^{10} (\hat{y}_i(x) - \bar{y}(x))^2. \tag{3.10}$$

To produce a single variance figure per mechanism, proportion, imputer, forecaster combination, this pointwise variance is aggregated across the full set of predictions in the horizon.

Bias can be computed in two ways: bias against the ground truth (the original test set), and bias against a baseline forecast. While both are computed, this research opts to focus on bias against a baseline. This baseline is a set of predicted values, one for each forecasting model, using models trained on the original, complete training set. Calculating bias against ground truth conflates the forecasting model's inherent errors with errors induced by the imputation pipeline. As the aim of this research is to analyse imputation bias propagation, by computing bias against this baseline, the imputation effects are isolated. In the formal definition, $y(x)$ is either the ground truth value, or the baseline forecast value, depending on the desired computation:

$$\text{bias}^2(x) = (\bar{y}(x) - y(x))^2. \tag{3.11}$$

As with variance, a single bias value is an aggregation across the set of predictions.

Finally, these OT values are inverse-transformed. This gives bias and variance as squared degrees Celsius, rather than the 0 to 1 scale used for reconstruction. However, hereafter, the squared bias values will simply be referred to as bias.

3.4 Ethical Issues

As discussed in Section 3.2.1, the dataset is publicly available and entirely non-sensitive. It does not contain any personal data, and was stored on university storage for the duration of this experiment. For this reason, no ethical concerns are raised. The ethics approval certificate is found in Appendix A.

CHAPTER 4

Results

In this chapter, key findings from EDA are presented (Section 4.1), before exploring the resulting synthetic missingness generation (Section 4.2). Data reconstruction analysis is then provided (Section 4.3) before focus turns to forecasting results (Section 4.4).

4.1 Exploratory Data Analysis

4.1.1 Target Variable Exploration

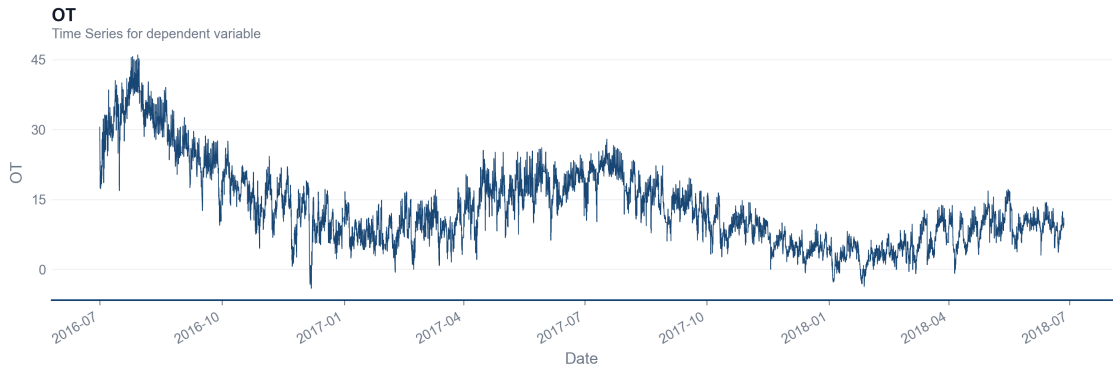


Figure 4.1. Time series for OT, the dependent variable, across full dataset.

Figure 4.1 shows the time series for the target variable, OT, across the whole dataset. There is a negative trend for the variable, as well as annual seasonality, with higher temperatures in the summer months. The monthly distributions, with peaks in the summer months, are shown more clearly in Figure 4.2. The values of OT are mostly positive, but the temperatures do fall below 0 at some points, ruling out multiplicative decomposition as a tool, and further down the pipeline, multiplicative smoothing components.

These temporal characteristics are further highlighted by the additive decomposition given in Figure 4.3, with this plot suggesting strong daily seasonality. Figure 4.4 emphasises this daily seasonality, with clear temperature spikes around the middle of the day, and

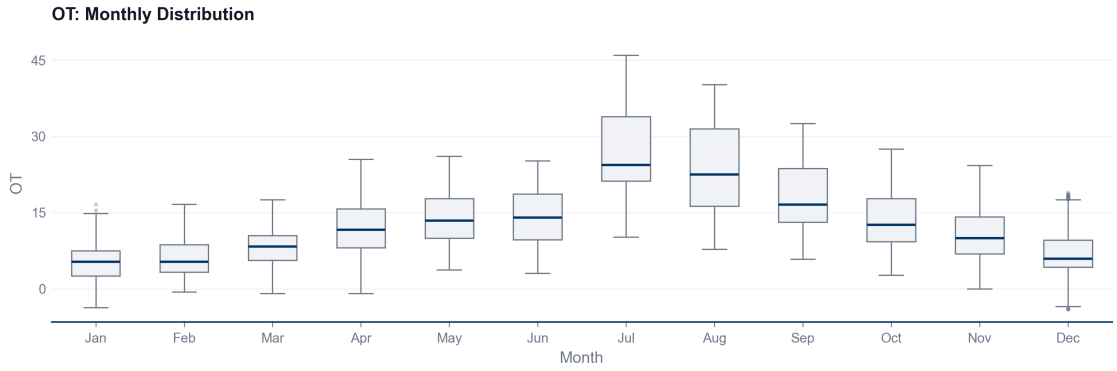


Figure 4.2. Monthly distributions for OT, showing annual seasonality with peak temperatures in the summer months.

drop offs overnight.

4.1.2 Predictor Variable Explorations

The TS for the predictor variables are given in Figures D.1, D.2, D.3, D.4, D.5, and D.6. They demonstrate similar temporal behaviours to OT in terms of the daily seasonal period, with some of the features displaying near identical patterns to each other across the TS. The correlation matrix in Figure 4.5 shows some extremely strongly correlated features, such as 0.99 between High Useful Load (HUFL) and Middle Useful Load (MUFL), and between High Useless Load (HULL) and Middle Useless Load (MULL). Notably, none of the predictor variables have any reasonably strong correlation with the target variable, with the highest value being only 0.22.

4.2 Synthetic Missingness Generation

Table 4.1 reports the actual missingness proportions for each of the missingness mechanisms at each desired missingness proportion, under a given random seed. As seen, the true rates are nearly identical in every case, with the discrepancies only being a consequence of integer rounding through the floor function. The actual missingness proportions are identical between mechanisms. This demonstrates that the mechanisms are not altering the proportions at all, as intended, but are instead only affecting which

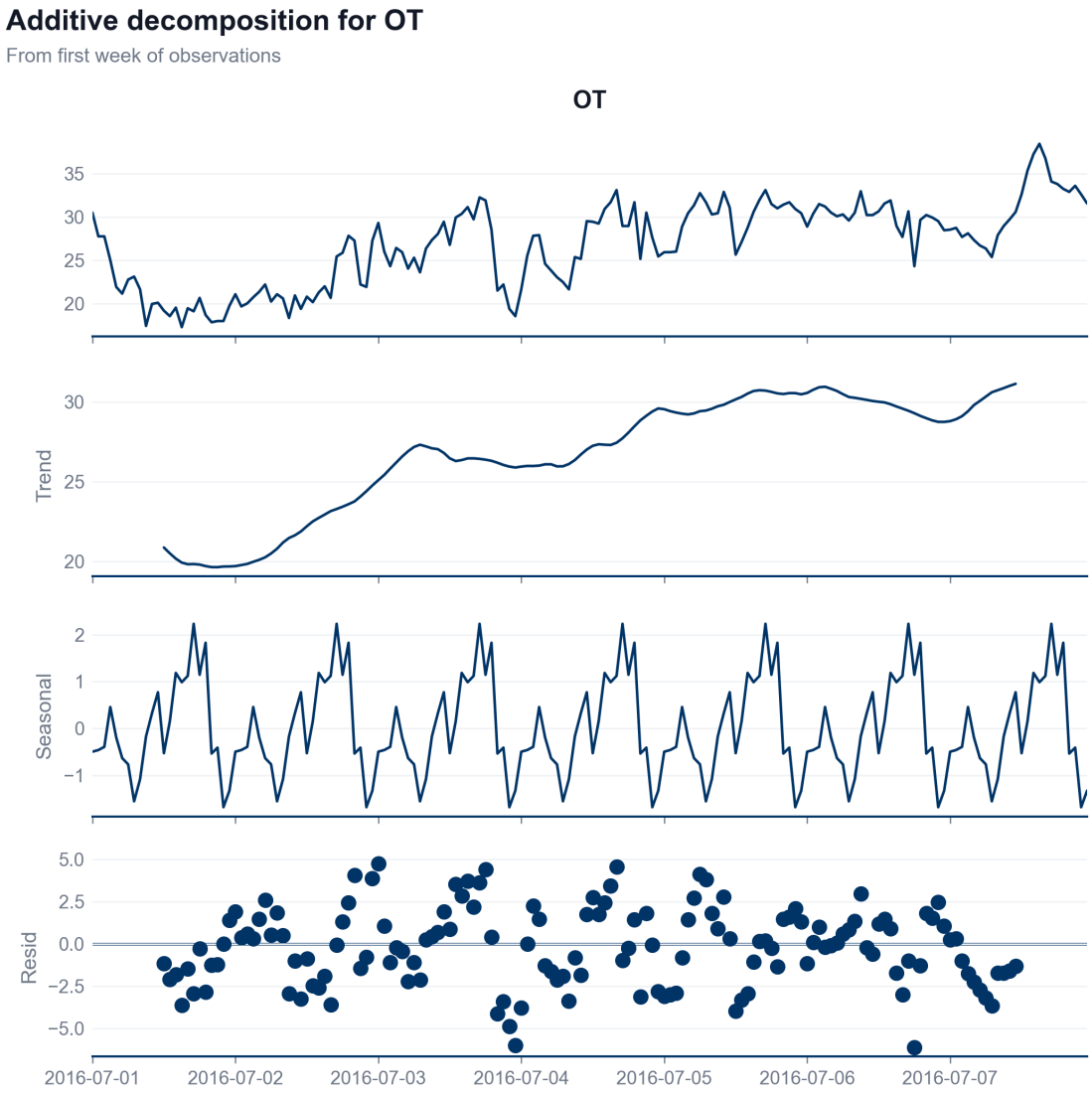


Figure 4.3. Additive decomposition of OT into trend, seasonal and residual components.

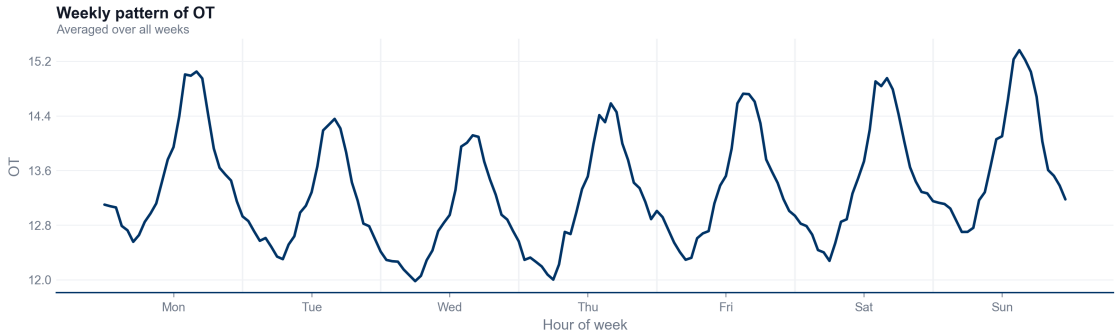


Figure 4.4. Mean OT across hours of the week, averaged across the full TS.

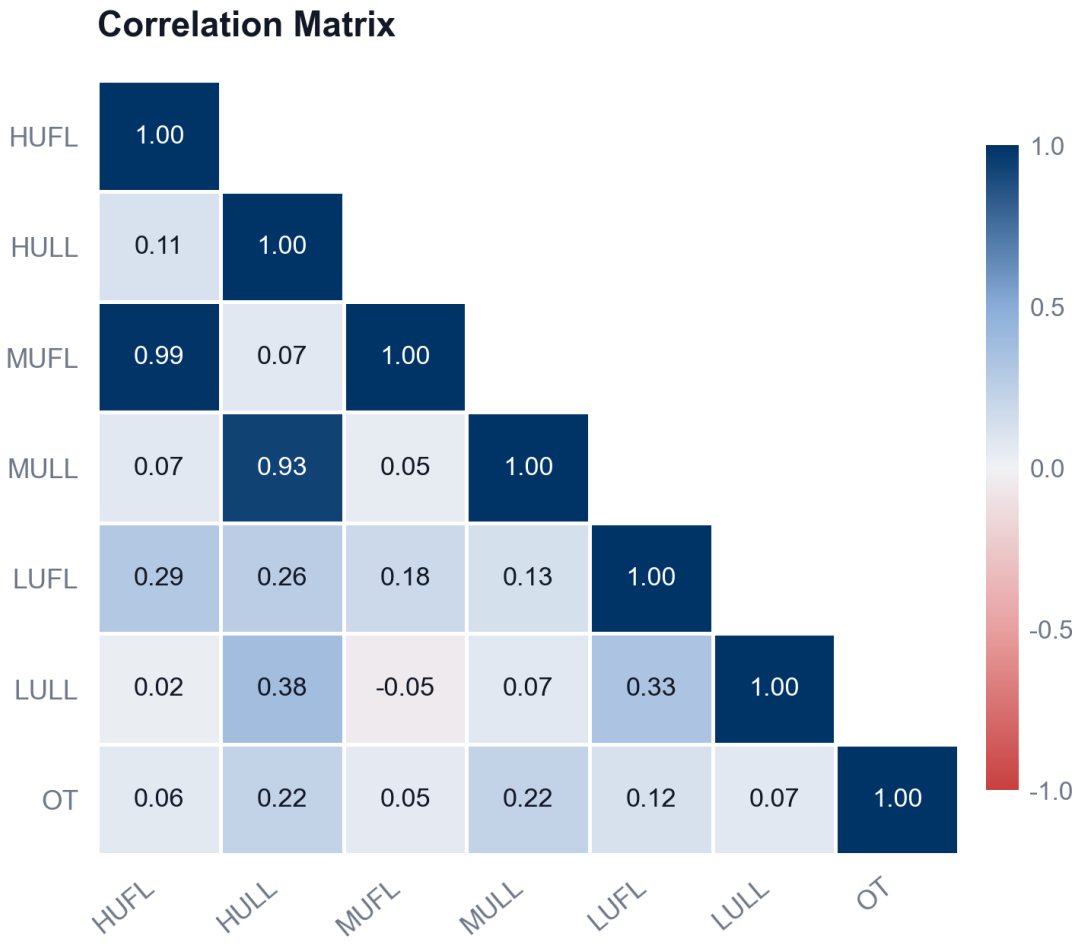


Figure 4.5. Correlation matrix for all features in the dataset, using Pearson correlation.

values are made to be missing, representing a consistent amputation methodology.

Table 4.1. True missingness proportions for each mechanism under random seed 1.

%	MAR	MCAR	MNAR
10	0.099991	0.099991	0.099991
30	0.299995	0.299995	0.299995
50	0.500000	0.500000	0.500000

To understand how the selected values change between mechanisms, it is possible to compare the distributions of the masked values (the values chosen to be missing), and the true, complete values, as in Figure 4.6 for OT. This shows the comparisons for a 50% missingness proportion, but the behaviour is consistent for other missingness proportions. The 10 and 30% proportion equivalents are given in Figures D.7 and D.8.

Under the MCAR mechanism, the distribution of the mask values is almost identical to the original feature's distribution, demonstrating that the value masking is truly random. As the size of the data increases, it is expected that the two distributions will converge given the uniform distribution of probability of missingness for MCAR, entirely independently of the values themselves. This pattern is further supported by the comparative distributions of HUFL under 30% MCAR in Figure 4.7, where, again, the distributions of the masked and true values are practically indistinguishable.

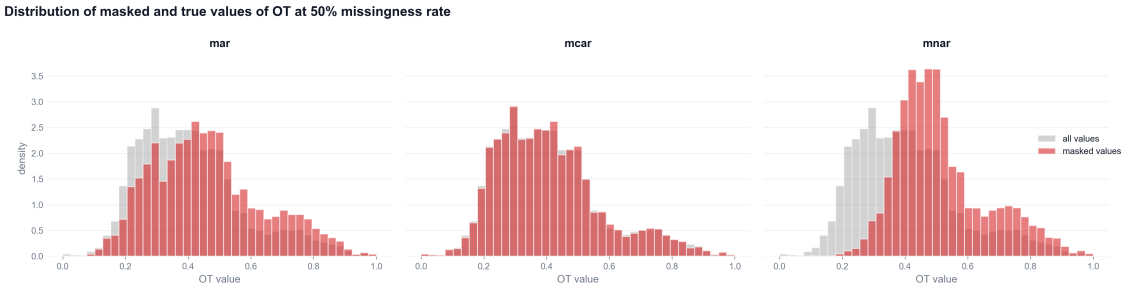


Figure 4.6. Comparison of masked and true value distributions of OT under each mechanism at a 50% missingness proportion.

As discussed in Section 3.1.1.2, synthetic MAR relies on a driver column to drive missingness within another feature. Figure 4.6 shows that the distribution of the masked OT values under MAR takes a similar shape to that of the MCAR distribution. OT's driver

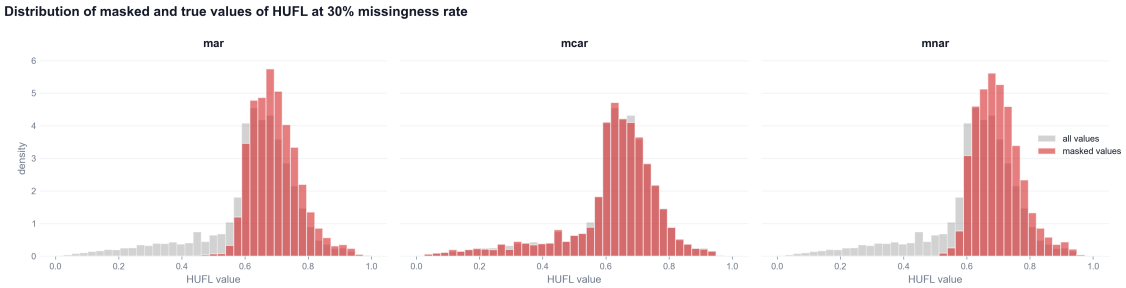


Figure 4.7. Comparison of masked and true value distributions of HUFL under each mechanism at a 30% missingness proportion.

is HULL, and Figure 4.8 shows how increases in the value of HULL correspond to a higher probability of missingness within OT. However, the correlation between HULL and OT, as seen in Figure 4.5, is very weak.

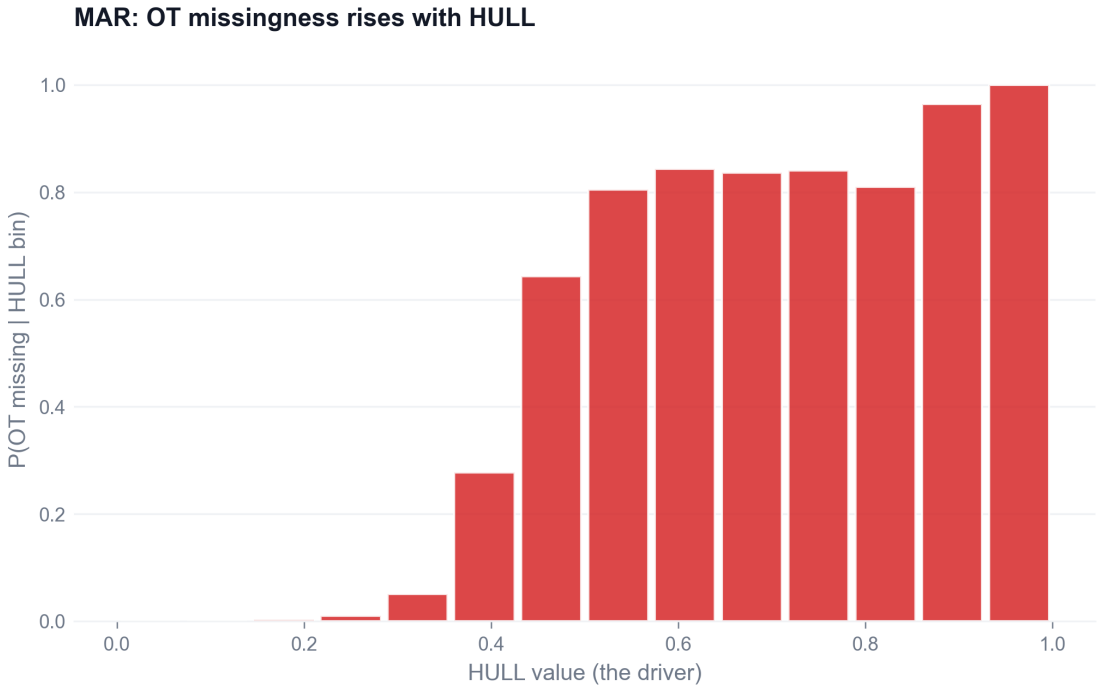


Figure 4.8. How the probability of missingness within OT under the MAR mechanism increases with the values of its driver column, HULL.

Contrast this with HUFL's masked MAR distribution in Figure 4.7. Here, the masked values are concentrated towards the higher values, leaving the tail unmasked. Additionally, the distribution is a very similar shape to that of the MNAR mechanism. This follows from HUFL having a 0.99 correlation with its driver feature, MUFL.

Recall that MNAR, in this research, uses the feature's own value to drive missing-

ness. Figure 4.9 shows how increased values of OT correspond to higher probabilities of missingness within the feature itself.



Figure 4.9. How the probability of missingness within OT under the MNAR mechanism increases with the values of itself.

Taken together, the mechanisms sit on a spectrum of how they are affected by values. MCAR sits at one extreme, where the value has no impact at all on missingness, with MNAR sitting at the other extreme, where the missingness is entirely dictated by the value itself. The probability of removal scales directly with the values being removed. MAR sits between these extremes, and where it sits is dependent on the correlation between the feature containing missingness and its driver. As seen in OT, a lower correlation with the driver leads to more MCAR-like behaviour, and a higher correlation tends towards MNAR. Furthermore, the plots show that while MCAR preserves distribution for the features, MAR and MNAR concentrate removal into value ranges, resulting in lower uniformity of missingness.

4.3 Reconstruction

The data for all reconstruction analysis is found in Table D.1, with figures in this section generated directly from these values.

4.3.1 Imputation Model Parameters

The imputers mean and LOCF carry no parameters. The parameters for the seasonal extension of ARIMA are given and justified in Table 4.2, with the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots provided in Figures D.9 and D.10, respectively. Figures D.11 and D.12 give the autocorrelations and partial autocorrelations after differencing by one season. The Augmented Dickey-Fuller (ADF) test (Dickey & Fuller, 1979) was used to test for stationarity.

Table 4.2. Identified ARIMA orders, applied identically to every column. Final model: ARIMA$(1,0,0)(0,1,1)_{24}$.

Parameter	Value	Component	Justification
d	0	Non-seasonal difference	ADF test $(p < 0.01)$
p	1	Non-seasonal AR	PACF cuts off at lag 1
q	0	Non-seasonal MA	Slowly decaying ACF
D	1	Seasonal difference	One difference at lag 24 removes the daily seasonality
P	0	Seasonal AR	Differenced PACF tails off
Q	1	Seasonal MA	ACF spike at lag 24 after differencing
s	24	Seasonal period	Daily seasonality

The remaining model parameters are given in Table 4.3. BRITS also exploits the daily seasonality, by setting the window length to be one season (24 hours).

4.3.2 Imputer Performance Comparison

Data for the reconstruction metrics are given in Table D.1. This section presents only the results for MAE, with RMSE plots in Figures D.13 and D.14 as the metrics show broadly similar patterns.

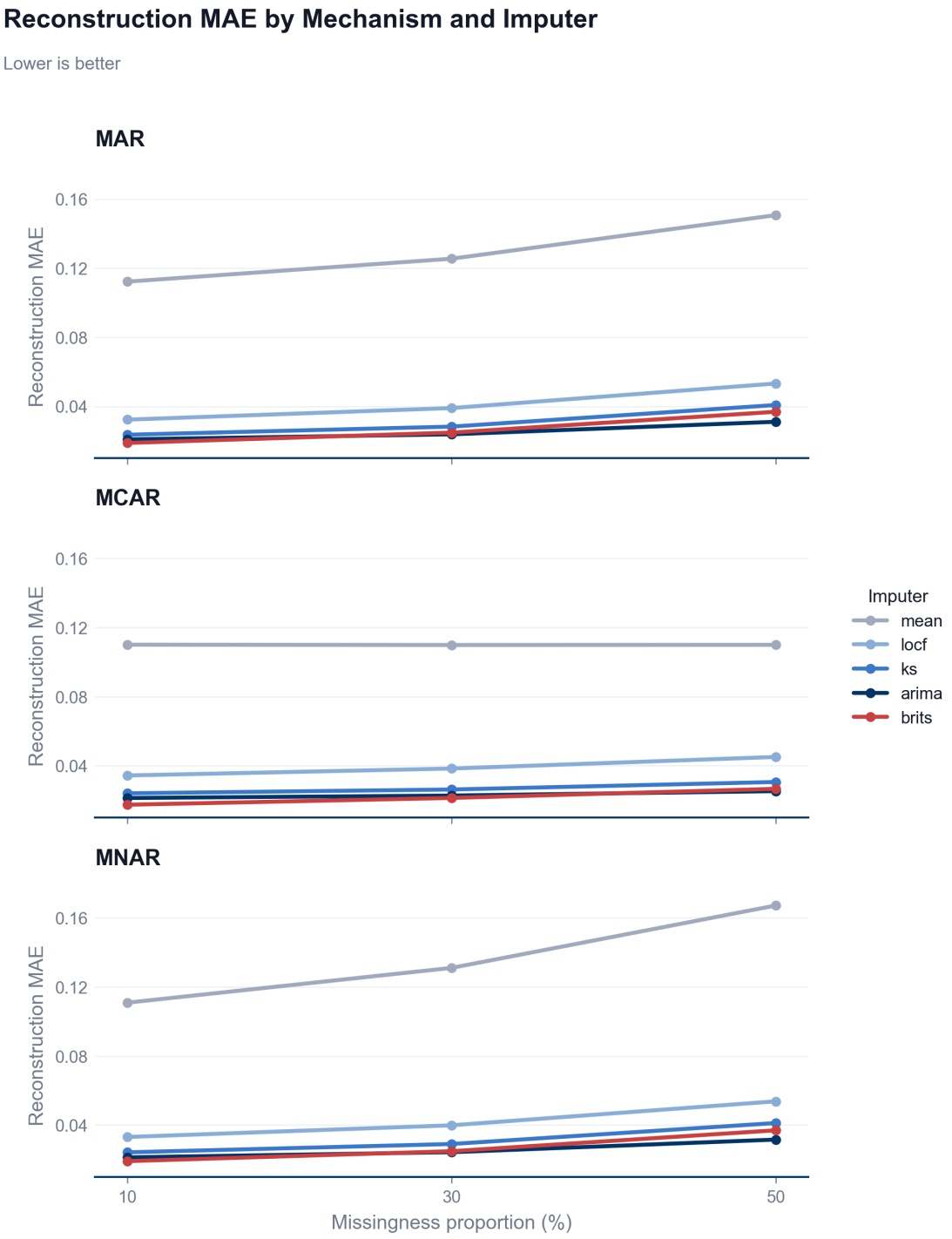


Figure 4.10. Comparison of MAE reconstruction by mechanisms and imputers, for varying missingness proportions.

Table 4.3. Parameter configurations of KS and BRITS imputers.

Imputer	Setting	Value
KS	Model	Local level
	Fitting	MLE, per column
BRITS	Window length	24 steps (one day)
	Hidden size	64
	Epochs	250
	Batch size	32

Figure 4.10 shows the aggregated MAE across each missingness mechanism, proportion and imputer. It is immediately obvious that mean imputation performs substantially worse than all other imputers across each mechanism. The MAE of the mean imputer is roughly between 2.5 and 3.5 times worse than the second worst performer, LOCF, in all cases.

Figure 4.11 provides the same data as Figure 4.10 but with the mean imputer excluded to give clearer insight. As expected, there is a clear trend across all imputers and mechanisms that higher missingness proportions lead to less accurate reconstruction.

The reconstruction performance in terms of ranking is broadly reflective of imputation model complexity. LOCF is the worst performing, followed by KSl ARIMA and BRITS perform similarly, but with subtle differences in how they are affected by missingness proportions.

At 10% missingness, BRITS outperformed ARIMA across each mechanism measured by MAE. At 30% missingness, their performance was very similar, with no clear winner. However, at the 50% missingness proportion, ARIMA consistently outperformed BRITS in all three mechanisms. While ARIMA's performance degrades with higher missingness proportions, as with all imputers, its degradation is far less substantial compared with that of BRITS. BRITS sees an immense performance drop off, particularly within MAR and MNAR.

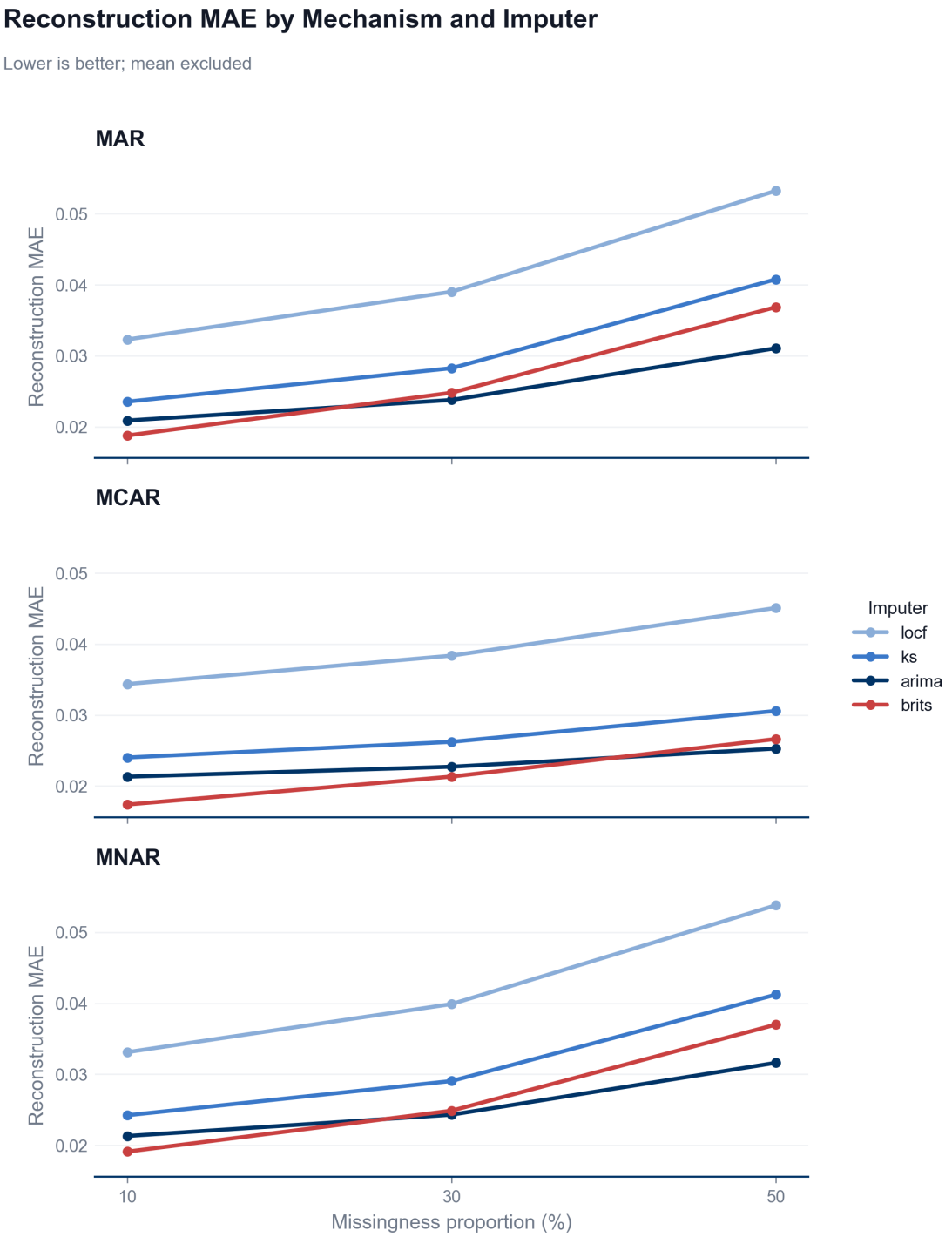


Figure 4.11. Comparison of MAE reconstruction by mechanisms and imputers, excluding mean, for varying missingness proportions.

4.3.3 Mechanism Comparison

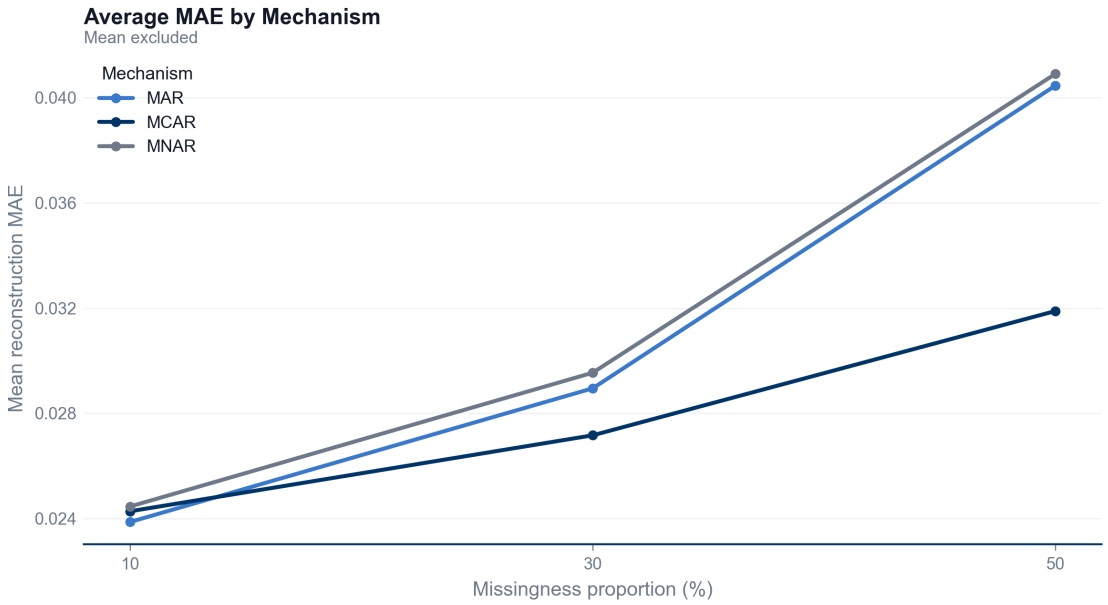


Figure 4.12. Mean MAE for each mechanism, averaged across all imputers excluding mean.

Figure 4.12 shows the average MAE across the three mechanisms and proportions by averaging across all the imputers, except mean. Again, mean is excluded here due to its drastically worsened performance that would skew results.

As the missingness proportion increases, MAR and MNAR degrade in performance very similarly. MCAR, however, demonstrates a less drastic performance drop off for the increased missingness proportions, shown by the shallower slope of its line in Figure 4.12.

4.3.4 Block Missingness

As first discussed in Section 3.1.1, block missingness is a temporal behaviour where consecutive values are missing. While the amputation methodology did not specifically target this phenomenon, an 11-hour block of missingness within OT did naturally occur under 50% missingness MCAR, shown in Figure 4.13.

This figure shows the behaviour of each of the imputers in reconstructing this missing data. The mean and LOCF imputers behave statically: they are frozen at the average value or last observed value, replacing each of the 11 missing points. The KS imputer

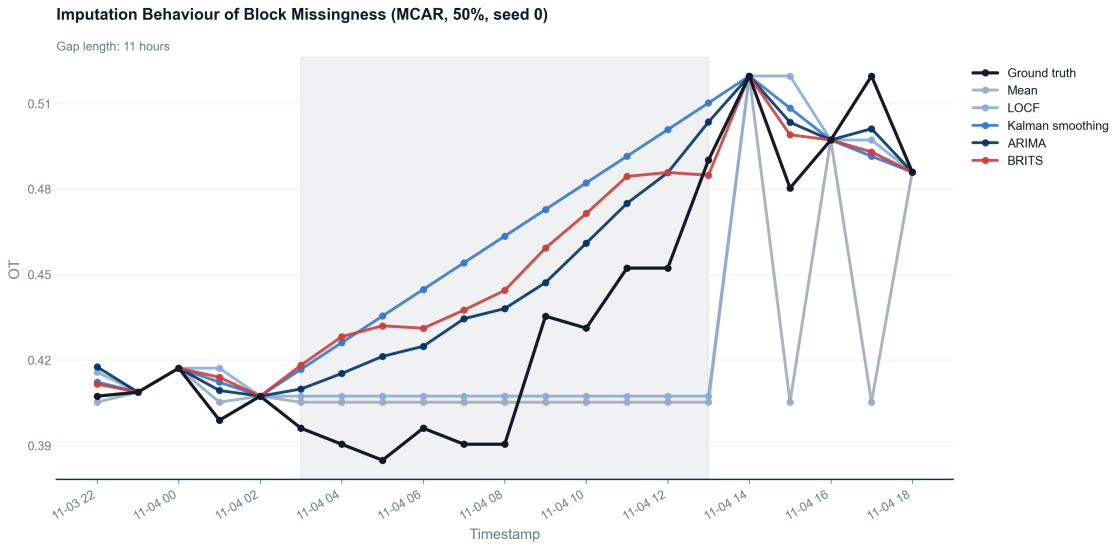


Figure 4.13. Imputation behaviour of imputation models for 11-hour block missingness within OT of the MCAR 50% Seed 0 dataset.

creates a smooth line of points across the missingness block, whereas ARIMA and BRITS are more variable in their filling, more similar to the inconsistency of the ground truth. The behaviour, and performance, of imputation methods in blocks of missingness offers room for further exploration.

4.4 Forecasting

4.4.1 Forecasting Model Parameters

Table 4.4. Parameter configuration of the ES forecaster.

Setting	Value	Justification
Seasonal period	24	Daily seasonality (Section 4.1.1)
Seasonality	Additive	Data non-positive under scaling
Trend	Additive/Damped	Long horizon

The parameters of the ES forecasting model are given in Table 4.4. The seasonal period is naturally 24, as this is the strongest seasonality, with additive methods being forced by the existence of non-positive values in the data, after min-max scaling.

The equivalent table for XGB is Table 4.5. Lagged features at lags 1, 24 and 168 are

Table 4.5. Parameter configuration of the XGB forecaster.

Setting	Value	Justification
Lags	[1, 24, 168]	Autocorrelation at lag 1, daily seasonality, horizon length
Calendar features	Month, day-of-week, hour	Encodes daily and longer calendar structure

used to reflect strong autocorrelation, seasonal patterns and horizon length. Calendar features of month, weekday and hour were further created. The remaining hyperparameters for XGB were left as default so that each model may be compared fairly. Additionally, tuning the hyperparameters of all 450 XGB models would be extremely computationally demanding.

As previously stated, but worth reinforcing, ES is univariate and is only trained on the OT feature that it is forecasting, compared with XGB being multivariate and exploiting feature-to-feature relationships.

4.4.2 Forecasting Results

The forecasting data, for all imputer, mechanism and proportion combinations are found in Table D.2 for XGB and Table D.3 for ES. The bias vs truth values are included for completeness.

4.4.2.1 Imputer Bias and Variance

Figure 4.14 compares the variance propagated into the forecasts by each imputer, averaged over all forecasts. Mean imputation carries the highest variance, surprisingly followed by BRITS. The variance then decreases as model complexity increases for LOCF, KS and ARIMA.

Figure 4.15 shows the aggregated bias for each imputer. While mean imputation propagated the most variance of the imputers, it was not by a huge margin. Here, mean propagates more than 5 times the bias of any other imputer; a similar structural finding to the reconstruction evaluation. Mean will, again, be excluded from further plots for this

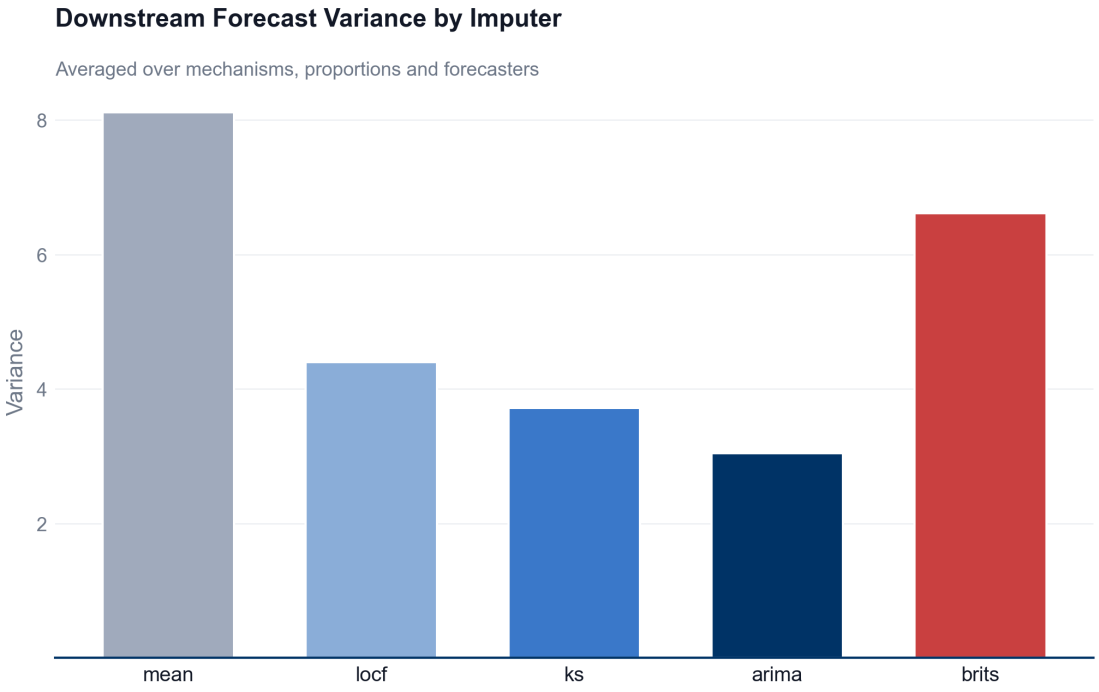


Figure 4.14. Comparison of average forecasting variance, by imputer, over a 168-hour forecasting horizon.

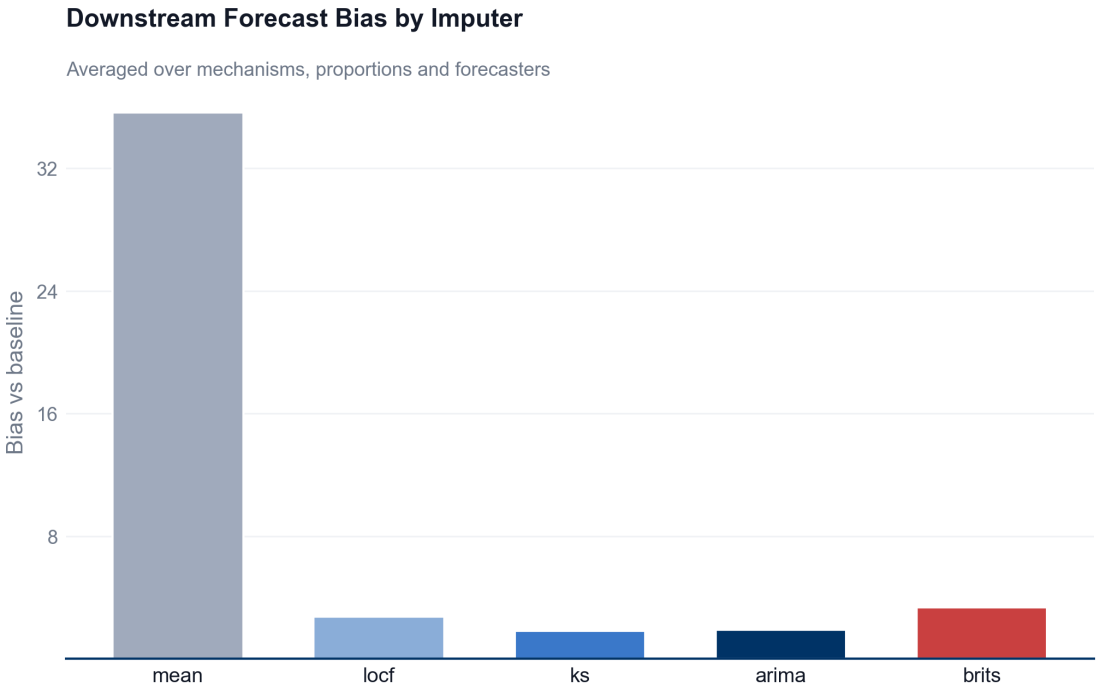


Figure 4.15. Comparison of average forecasting bias^2, by imputer, against baseline models over a 168-hour forecasting horizon.

reason. Also as with variance, BRITS has the second highest bias, with marginally less bias for LOCF, and with KS and ARIMA showing similar bias levels in the forecasts.

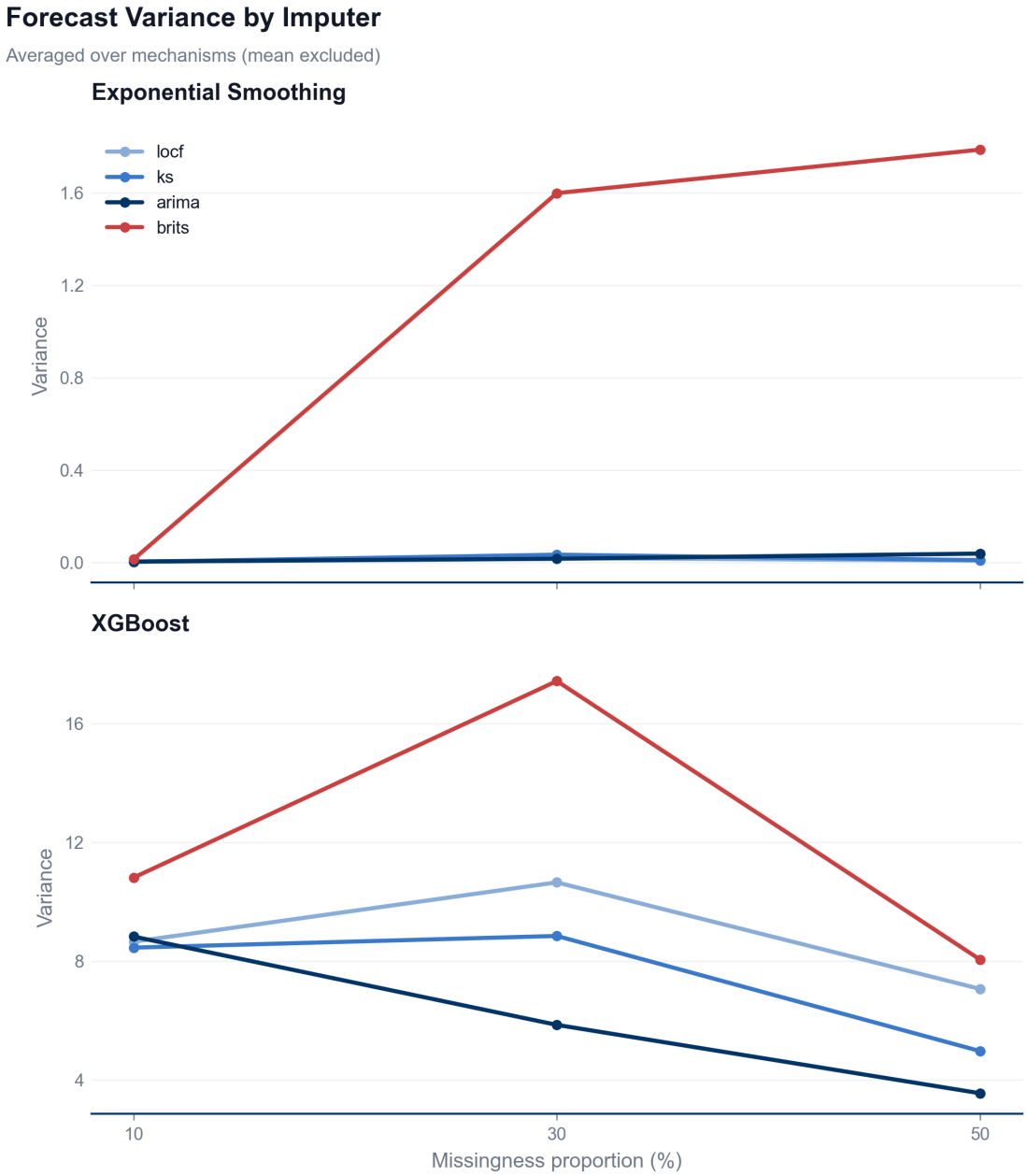


Figure 4.16. Comparison of forecasting variance by imputer, separated by forecaster, over a 168-hour forecasting horizon.

Separating by forecasting model, and into different missingness proportions, as in Figure 4.16, shows that at a low missingness proportion, each imputer is similar in the propagation of variance into the forecast for both forecasting models. For ES, for all imputers other than BRITS, the missingness proportion only negligibly increases the

downstream variance, if at all. For XGB the pattern is less clear. There appears to be a trend of decreasing variance for 50% missingness, but with a small spike, most notably within BRITS, of high variance under 30% missingness.

The bias equivalent, Figure 4.17, shows a very similar pattern within ES. BRITS increases bias for higher missingness proportions, while the remaining imputers are very consistent with the bias propagation. All of these imputers carry much higher bias within XGB, but the spread of bias across the proportions is inconsistent: BRITS and KS remain at a fairly constant bias propagation for the proportions, whereas ARIMA's bias drops drastically at 30%, and LOCF sees an increase.

4.4.2.2 Missingness Mechanism Bias and Variance

Analysing the mechanisms rather than the imputers, Figure 4.18 shows that MAR strongly affects the variance propagated into ES forecasts for 30% and 50% missingness proportions. Each mechanism propagates little variance at lower missingness levels, with MNAR almost static across all levels. XGB demonstrates low variance for all mechanisms at 50% missingness, with MAR and MCAR decreasing variance with higher proportions. MNAR sees a spike of variance within XGB forecasts at 30% missingness.

The bias by mechanism, Figure 4.19, for ES takes a very similar pattern to the variance. MAR sees high bias for higher missingness, with the other mechanisms staying relatively stable with bias propagation. XGB sees decreasing bias for higher proportions for both MAR and MNAR, with a reverse of increasing bias for MCAR.

4.4.2.3 Forecaster Bias and Variance

Figure 4.20 summarises the bias-variance behaviours of the two forecasting methods. ES exhibits both low bias and variance, with similar levels for both. XGB has much higher values for both bias and variance compared with ES, and sees proportionally far higher variance than bias.

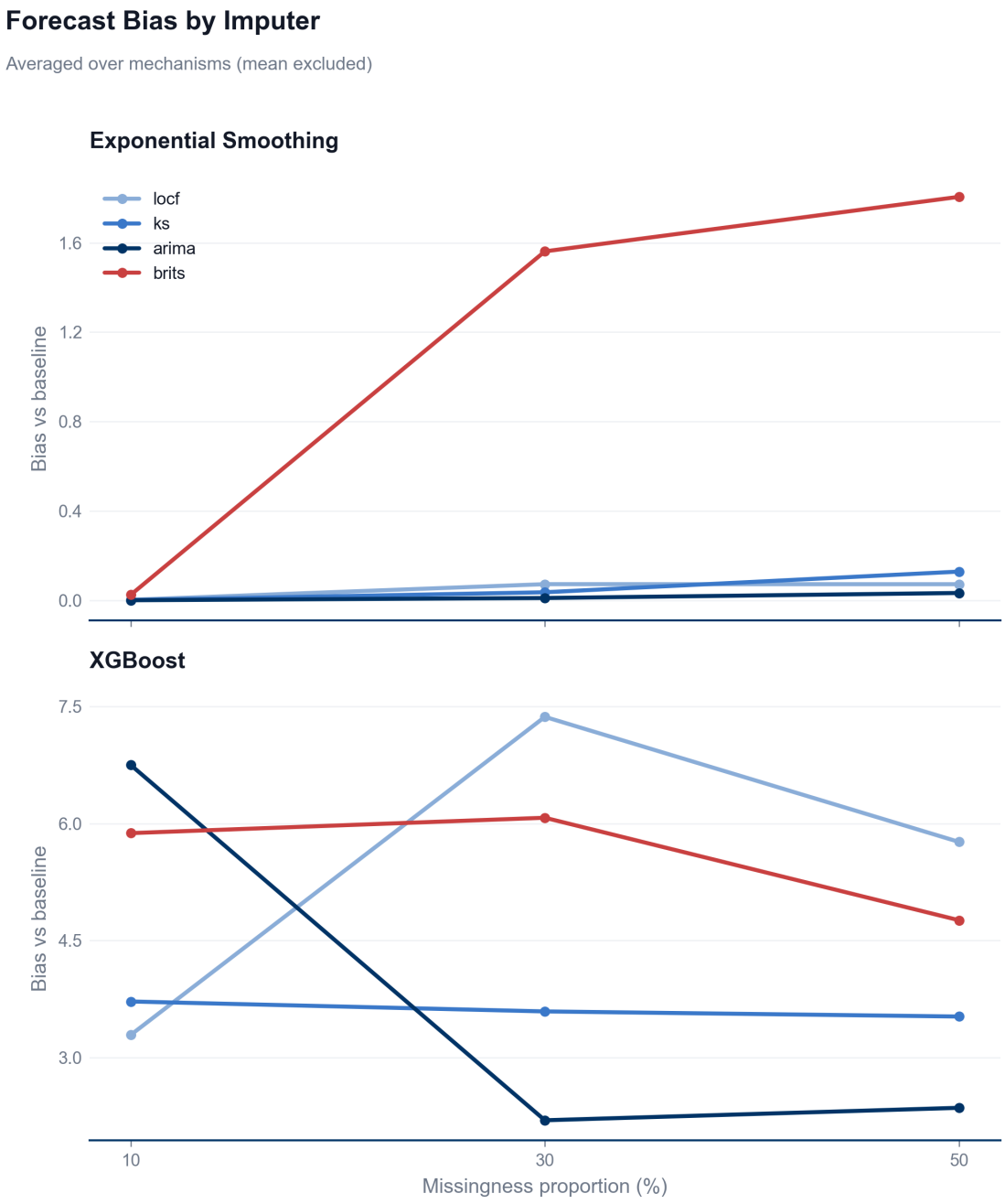


Figure 4.17. Comparison of forecasting bias by imputer, separated by forecaster, over a 168-hour forecasting horizon.

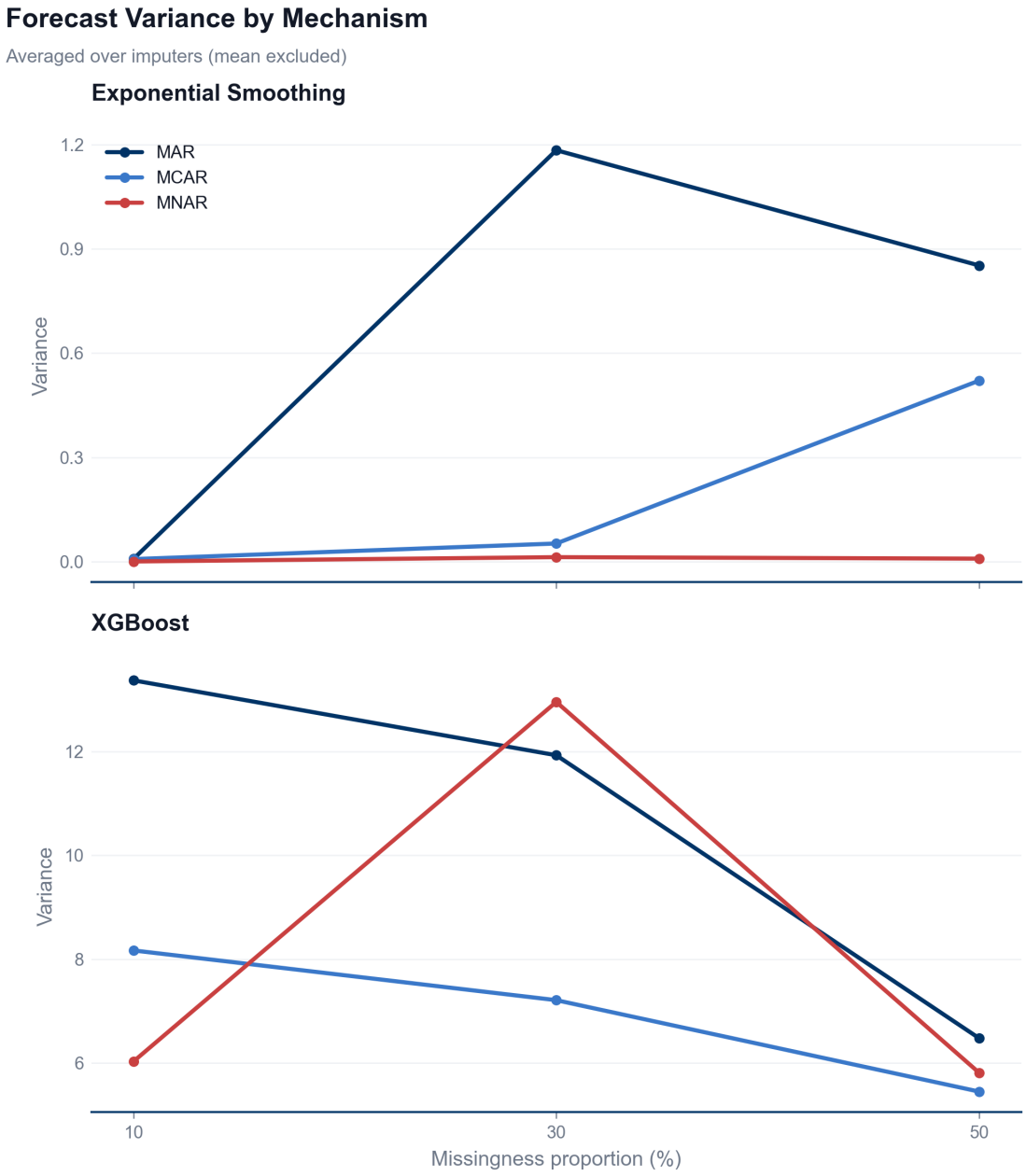


Figure 4.18. Comparison of forecasting variance, by missingness mechanism, over a 168-hour forecasting horizon.

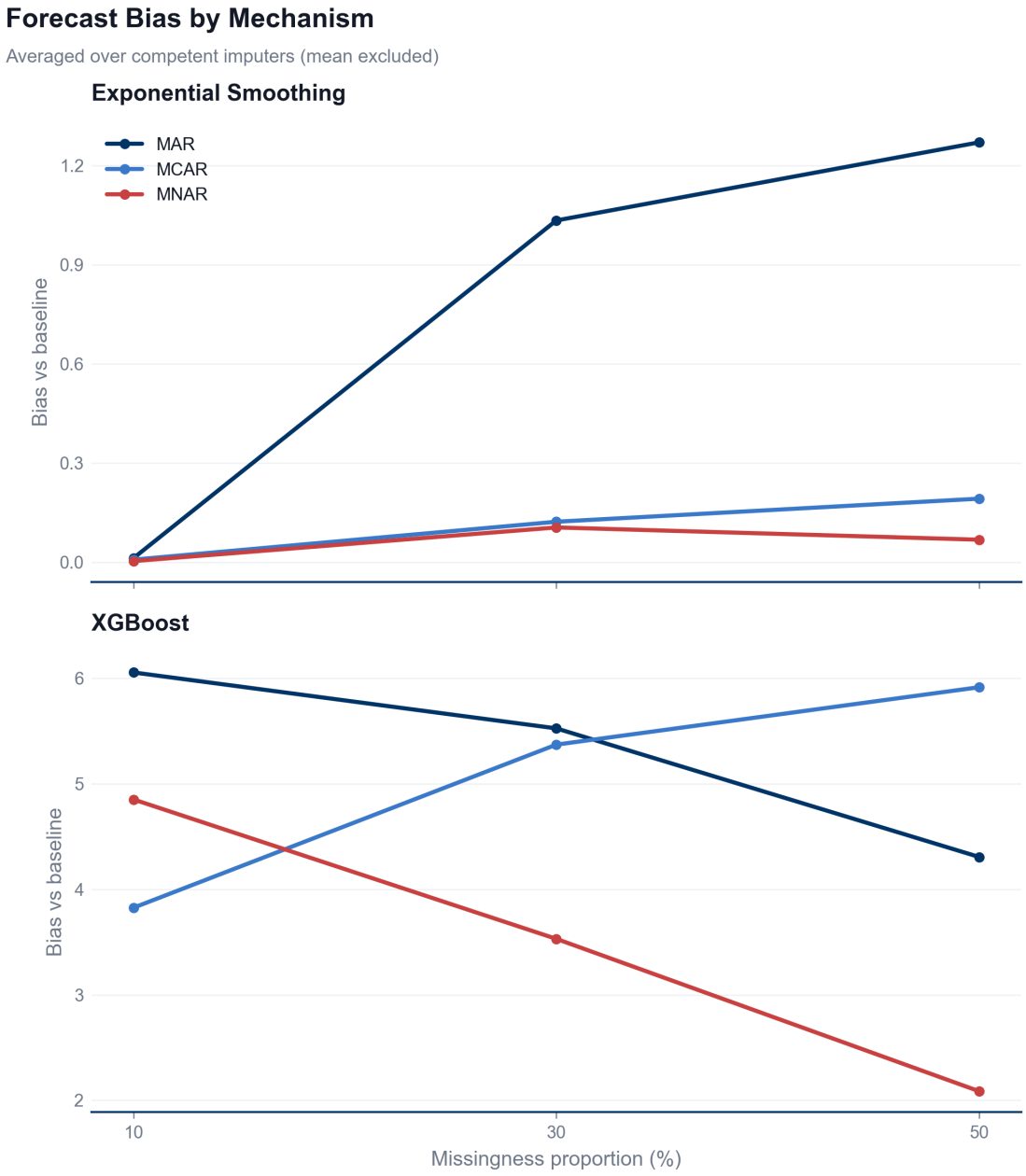


Figure 4.19. Comparison of forecasting bias[2], by missingness mechanism, against baseline models over a 168-hour forecasting horizon.

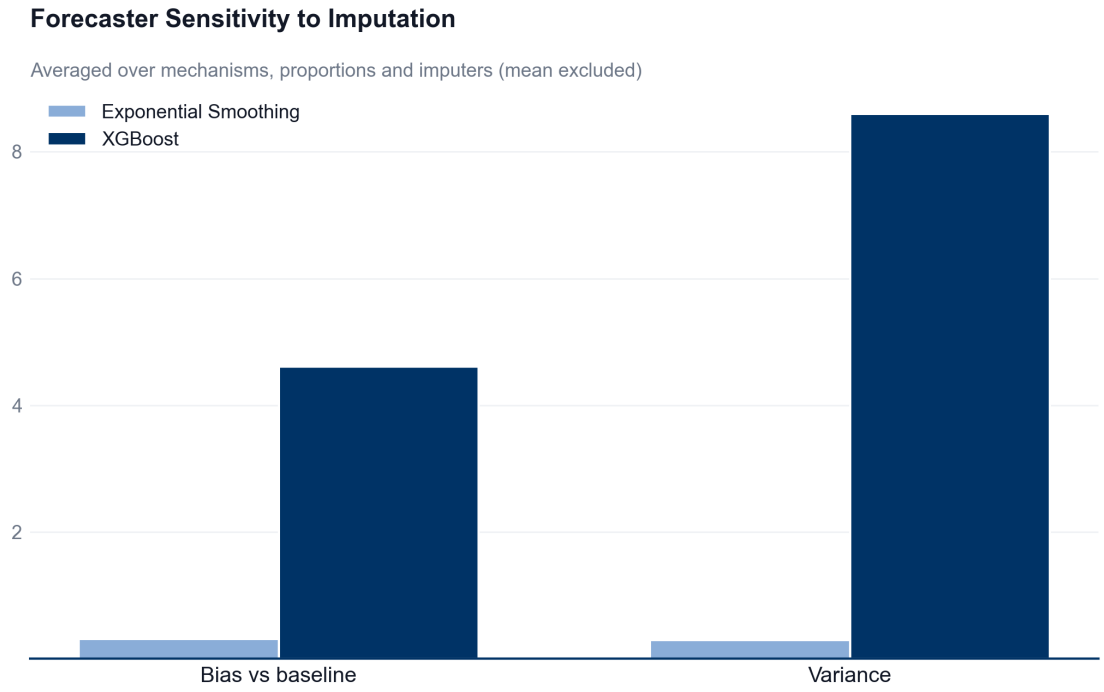


Figure 4.20. Overall comparison of bias, against baseline models, and variance, by forecaster over a 168-hour forecasting horizon.

CHAPTER 5

Discussion

This chapter aims to interpret and analyse the results of both the reconstruction (Section 5.1) and forecasting (Section 5.2) components of the experiment. Implications of these findings are then discussed (Section 5.3).

5.1 Reconstruction

Section 4.3 has a clear overarching theme in terms of imputation model and reconstruction accuracy: more sophisticated models offer superior data reconstruction performance. Mean imputation, the most basic method, demonstrated extremely poor reconstruction performance, compared to the other imputation models. LOCF offered strong performance given its simplicity; this is likely a consequence of the high autocorrelation within the features, where consecutive values showed strong relationships. Figure 4.13 showed, however, that for larger blocks of missingness, LOCF suffers from a frozen observation, suggesting that for higher missingness proportions, or where blocks of missingness are frequent, it is not the optimal method, despite its low computational cost. KS outperformed both naïve methods, showing it is successfully leveraging temporal dependencies (Kannegowda et al., 2024; Saputra et al., 2021), with ARIMA and BRITS offering comparable performance with reconstruction on the whole. However, at lower missingness proportions, measured by MAE, BRITS was consistently the strongest imputer, but was overtaken by ARIMA at higher missingness proportions. As a DL model, BRITS requires a lot of data; the dataset only contains 17,420 observations when complete, thus splitting into 70% training data and removing 50% of the data within these observations leaves BRITS lacking in data to learn from. ARIMA's orders are fixed, and thus it does not require abundant data to learn the seasonal patterns, whereas BRITS must learn these patterns. BRITS offering the strongest performance is in agreement with the literature (Cao et al.,

2018), up to where it suffers from higher missingness proportions. Kazijevs and Samad (2023) found that no single imputer dominated across multiple datasets, indicating that performance rankings are dataset-dependent. In this case, ARIMA outperforming BRITS at higher missingness levels may be a consequence of the limited data.

Regarding mechanisms, this research has directly addressed the issue raised by Y. Zhou et al. (2024), in that research is dominated by MCAR. Section 4.3.3 demonstrated that MCAR is least affected by higher missingness proportions, for the mechanism implementations used in this research. This is a consequence of how missingness is distributed through each mechanism. Figures 4.6 and 4.7 showed that MCAR does not change the feature distributions even under high missingness proportions, whereas MNAR strips upper values through the value itself, and MAR removes them through driver correlation, aligning with Oberman and Vink (2024). By removing value ranges, MAR and MNAR leave imputers to extrapolate blindly, compared with MCAR where imputers can interpolate within the full data range. With strong seasonality in the data, it is likely that removing value ranges causes block missingness, leading to further imputation errors.

5.2 Forecasting

Figure 4.20 showed that XGB carried much higher bias and variance than ES, with ES having fairly balanced bias and variance, while XGB was dominated by variance. Having the same reconstructed data yielding low variance in ES yet substantial variance in XGB indicates that it is the forecaster, rather than the imputation process, that dictates how reconstruction error surfaces in downstream tasks. As a multivariate forecaster, XGB is exposed to imputation errors across all features, compared with ES that only saw errors in OT. Furthermore, the XGB model was implemented using a recursive approach, inflating errors by accumulating error over the horizon. The recursion accounts for the bias, agreeing with Ben Taieb and Atiya (2016), with the high variance suggesting overfitting. The variance is the spread of forecast values across each of the 10 reconstructions of each configuration, and XGB fits each of these reconstructions closely, as a flexible tree

ensemble, resulting in highly varied predictions. This is precisely how James et al. (2013) describe the variance component; ES is fit only on OT, with forecasts changing little between reconstructions, resulting in low variance.

While BRITS was the best performing imputer at a 10% missingness proportion (under MAE), and best or second best at 30 and 50% proportions (Section 4.3.2), excluding mean, it demonstrated the highest variance propagation in both the ES and XGB forecasts (Figure 4.16), by far the highest bias propagation into ES forecasts, and the second highest bias propagation at each missingness proportion of the XGB forecasts (Figure 4.17). This suggests that strong reconstruction metrics do not prevent error propagation into forecasts, and that optimal, in terms of MAE and RMSE, reconstruction does not result in optimal forecasts. Using the same dataset, Du et al. (2024) also concluded that imputer choice measurably propagates into forecasts. Their focus was on MAE within forecasts; their results showed that their strongest imputer (SAITS) did not result in the best forecasts, whereas this research has shown that optimal reconstruction metrics can induce forecast instability, beyond pure forecasting performance. Kazijevs and Samad (2023) argue for the need for downstream evaluation of imputation; this research supports this need and extends it to highlighting that broad metrics like MAE mask forecast instability induced through imputation errors.

An unexpected result was the non-monotonic forecast variance behaviour of XGB across missingness proportions. The XGB panel in Figure 4.18 showed decreasing variance for both MAR and MCAR with increasing missingness proportions, while MNAR exhibited a variance spike at 30% missingness. This pattern was most exaggerated through BRITS (Figure 4.16). A plausible explanation for this behaviour is in how variance is measured as a spread of 10 forecasts from reconstruction configurations, under different seeds. Common to all mechanisms is that at 50%, the imputers, BRITS above all, have little data to learn from, thus imputed values are more homogeneous across seeds and the spread of forecasts narrows. This accounts for MAR and MCAR directly, with their highest variances being at 10% missingness and steadily declining as the homogenisation develops. By contrast, MNAR and BRITS exhibit a spike at 30% missingness before

the 50% decrease. At 30%, there is significant disparity between the reconstructions as there is a large proportion of data to be filled, but imputers are exposed to different remaining patterns to reconstruct from. This different learning surfaces in the forecasts. MNAR amplifies this peak by specifically removing upper tails, leaving extreme values with no local observations and thus highly variable reconstructions across seeds. This is speculative, however, and this result offers scope for further exploration, but it is consistent with BRITS being sensitive to missingness proportions.

5.3 Implications

The clearest implication from this research concerns how imputation methods should be chosen and evaluated. When forecasting is the downstream task, reconstruction accuracy metrics alone are an insufficient basis on which to choose an imputer: reconstruction accuracy ranking does not demonstrate how instability is propagated into forecasts. A bias-variance decomposition of forecasts highlights instability and should accompany reconstruction and forecasting accuracy metrics in imputation analysis.

A secondary implication regards the pairing of imputation and forecasting models. The extent to which imputation errors were propagated into forecasts was dictated by the forecasting models themselves: ES was largely insensitive to imputation errors as its univariate nature only exposed it to errors within the target feature, whereas XGB being multivariate and recursively built amplified the same reconstruction errors into much greater bias and variance. Volatile imputers are penalised more heavily by multivariate or recursive forecasters. Thus, it is recommended that imputers and forecasters should be considered as a pair in these contexts.

A third implication concerns the nature of the missingness. At low missingness proportions, BRITS was the strongest imputer but was overtaken by ARIMA at higher proportions. The computational demand of DL imputation methods like BRITS is repaid only where there is sufficient data to be trained on. This suggests that statistical methods, such as ARIMA and KS, are safer at higher missingness proportions, as they are able

to capture temporal patterns more readily, reserving DL imputers for instances when missingness proportions are low and data is plentiful.

CHAPTER 6

Conclusions

6.1 Answers to the Research Questions

This research designed and executed a systematic framework for comparing time series imputation methods in reconstruction performance, and how they propagate systematic error and instability into downstream forecasting tasks, under synthetic missingness at varying proportions. The research questions that this study aimed to answer, first stated in Section 1.2.2, are restated here and answered.

1. How do different missingness mechanisms (MAR, MCAR, MNAR) and missingness proportions affect reconstruction performance of time series imputation methods?

   Reconstruction accuracy degrades as missingness proportions increase for all imputers and missingness mechanisms. The rate of degradation is determined by the mechanism, with MCAR least affected and preserving data distributions, while MAR and MNAR suffered from blind extrapolation after stripping value ranges. Reconstruction accuracy scaled with imputation model sophistication; mean imputation in particular demonstrated very low performance, while the DL accuracy advantage was subject to the presence of sufficient observed data, with BRITS overtaken by ARIMA at higher missingness proportions.

2. To what extent do different time series imputation techniques demonstrate different bias-variance trade-offs within downstream forecasting tasks?

   While different missingness mechanisms and imputation models exhibited unique bias-variance profiles within forecasting tasks, the forecasting model itself was largely the dictating factor in error characteristics. Forecasting model architecture, such as uni- or multivariate, ruled how imputation errors were allowed to propagate

into the forecasts. Furthermore, the best reconstruction accuracy did not result in optimal forecasting, with the strongest reconstructor, evaluated by MAE, propagating the most instability into forecasts of the competitive imputation models.

6.2 Limitations and Future Work

Several limitations restrict this work, each scoping a direction for future research:

1. **Single dataset** The experiment was conducted using only a single dataset, within one domain. While the dataset grounds findings in sensor data, replication across a range of datasets, each containing different temporal behaviours, is required before generalising conclusions.

2. **Single imputation only** Multiple imputation methods, such as MICE, were not included. Extending the imputation methods to include these and profiling their error propagations into forecasts would widen the comparison.

3. **No temporal missingness patterns** Missingness was generated on a cell-wise basis, without accounting for the temporal ordering of values and thus neglecting time series specific missingness behaviours. Modelling temporal patterns such as block missingness is a natural extension.

4. **Lack of tuning** The ARIMA model used the same orders across each column, and the forecasters ran on default hyperparameters. Specific configuration of these models would tighten error estimates.

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APPENDIX D

Supplementary Figures and Tables

Figures

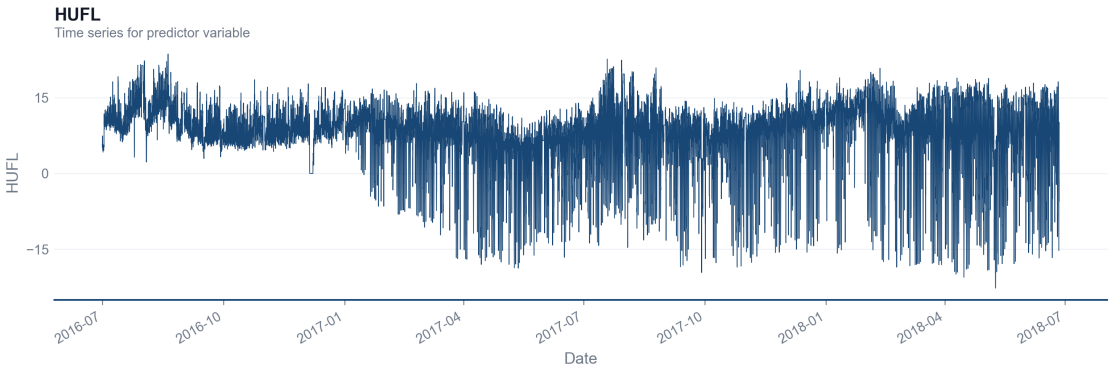


Figure D.1. HUFL time series across full dataset.

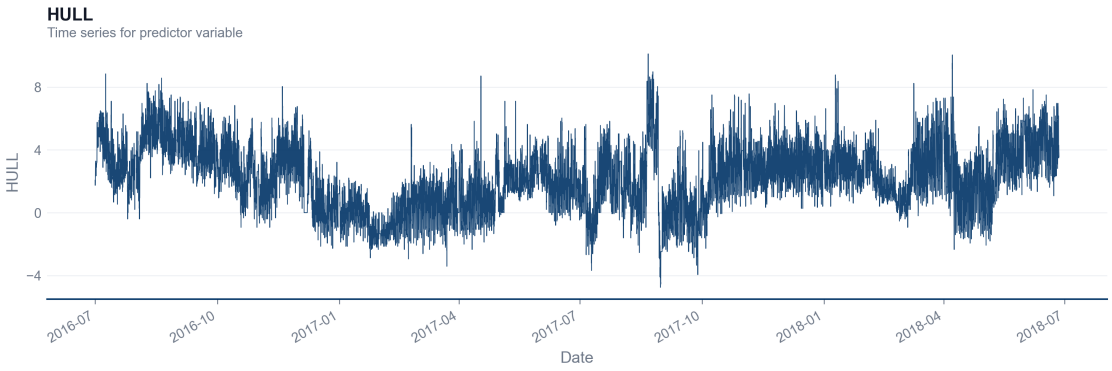


Figure D.2. HULL time series across full dataset.

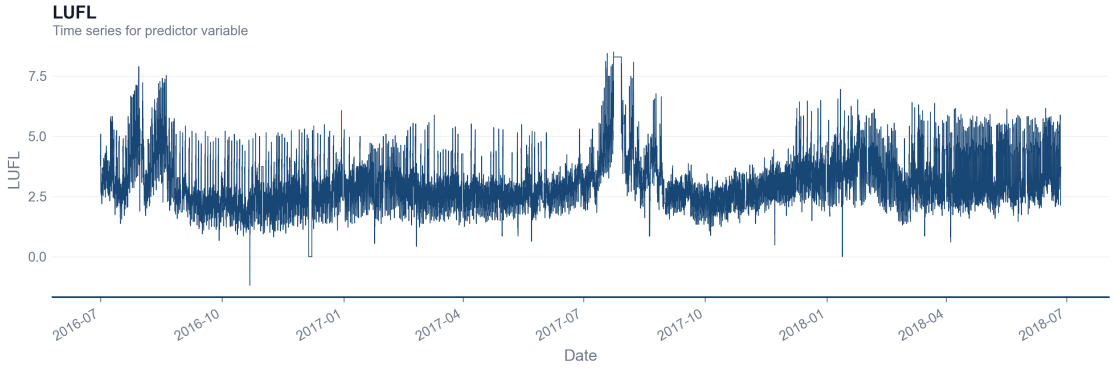


Figure D.3. LUFL time series across full dataset.

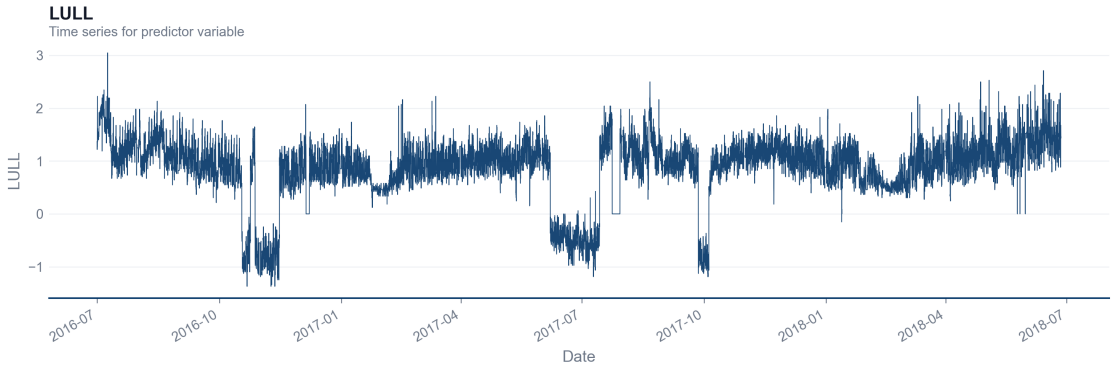


Figure D.4. LULL time series across full dataset.

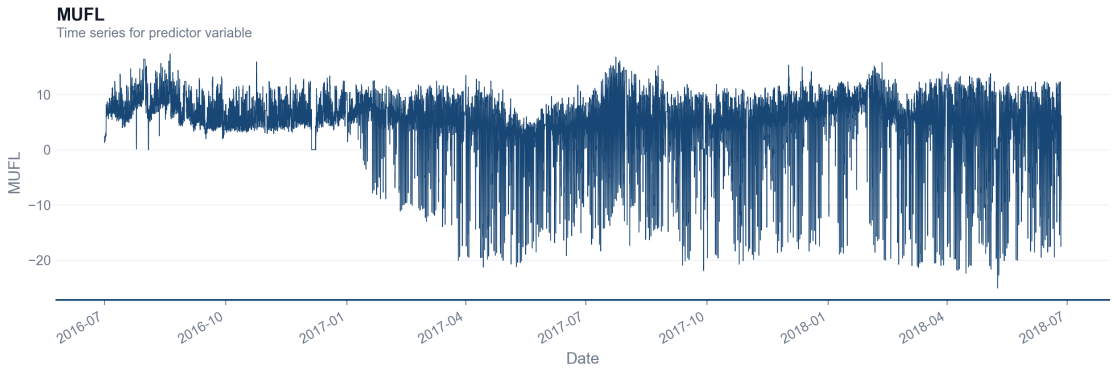


Figure D.5. MUFL time series across full dataset.

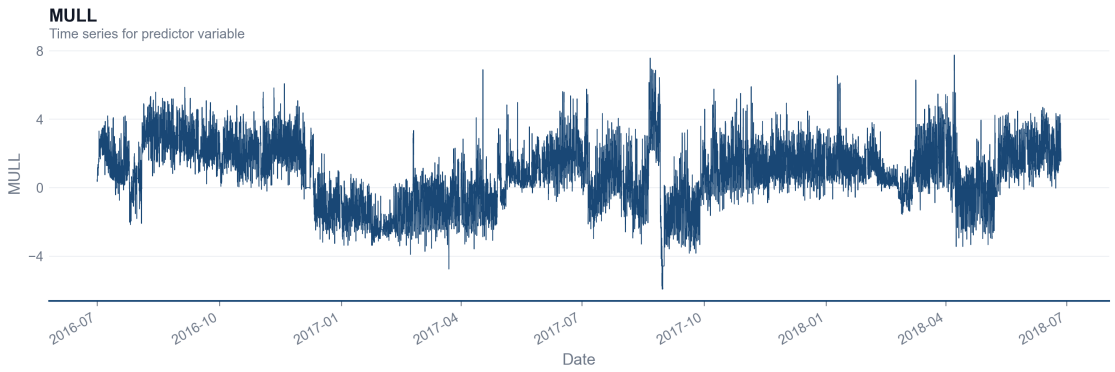


Figure D.6. MULL time series across full dataset.

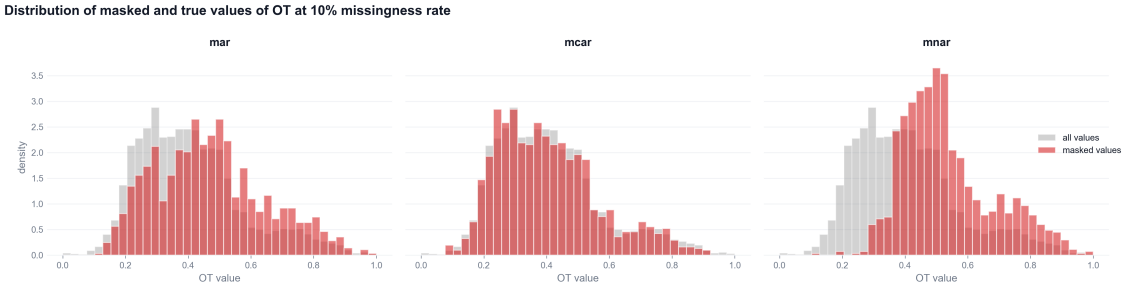


Figure D.7. Comparison of masked and true value distributions of OT under each mechanism at a 10% missingness proportion.

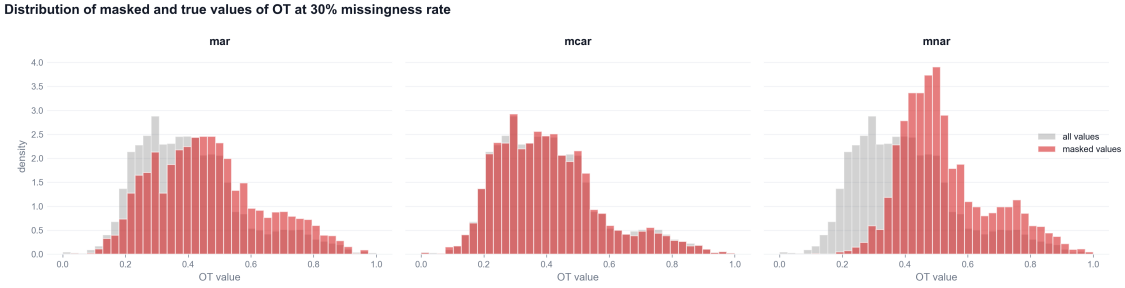


Figure D.8. Comparison of masked and true value distributions of OT under each mechanism at a 30% missingness proportion.

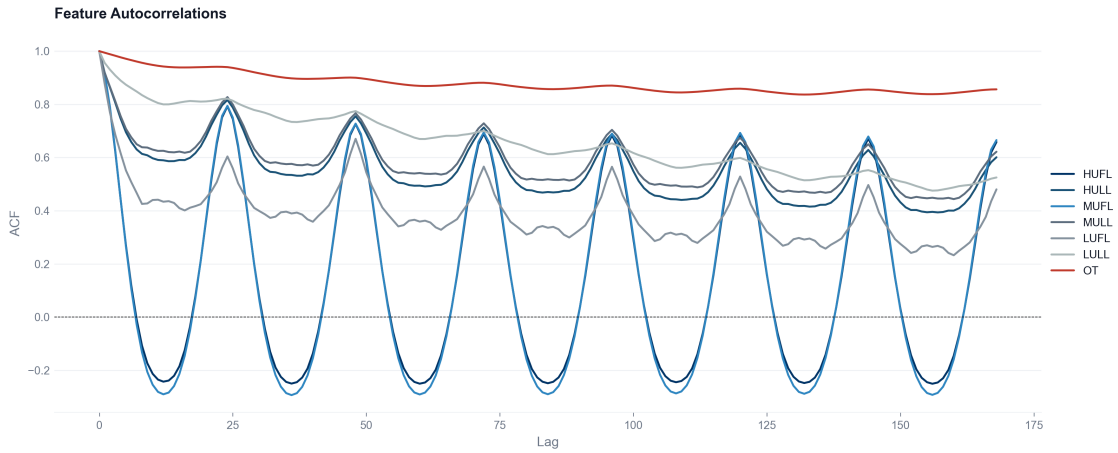


Figure D.9. Feature autocorrelations for all features in the dataset, up to lag 168.

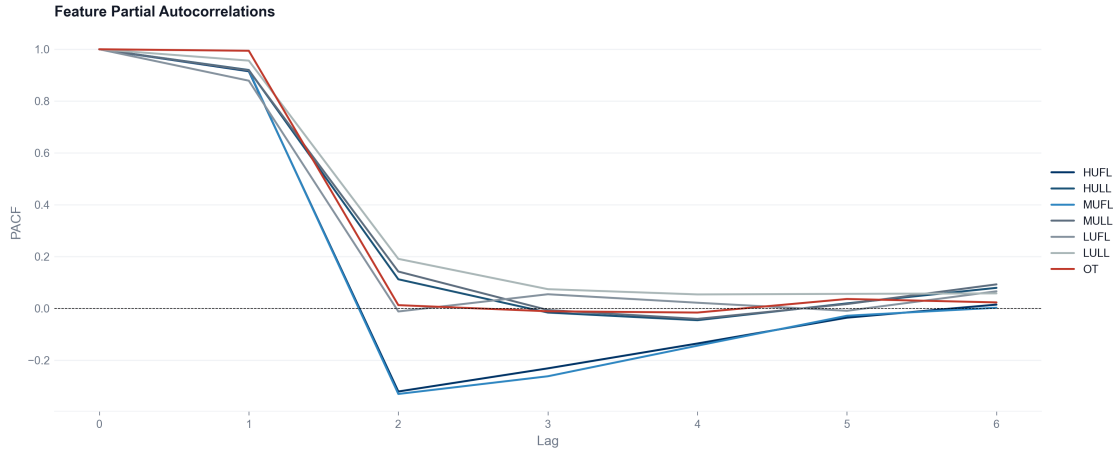


Figure D.10. Feature partial autocorrelations for all features in the dataset, up to lag 6.

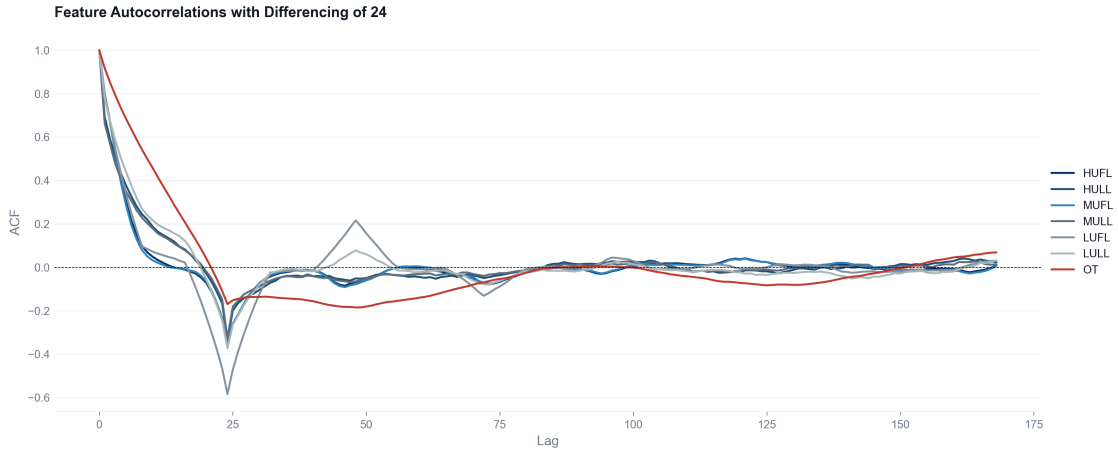


Figure D.11. Feature autocorrelations for all features in the dataset, up to lag 168, with 24-hour differencing.

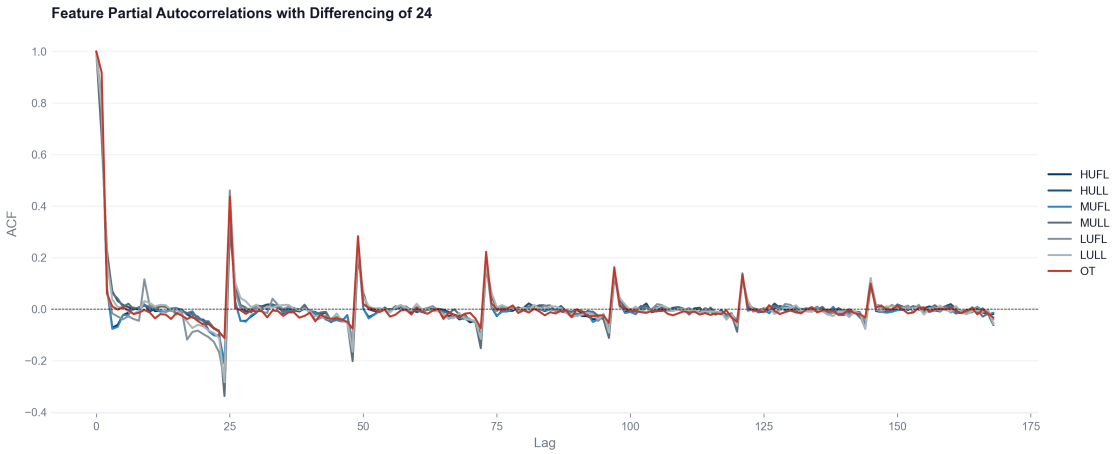


Figure D.12. Feature partial autocorrelations for all features in the dataset, up to lag 168, with 24-hour differencing.

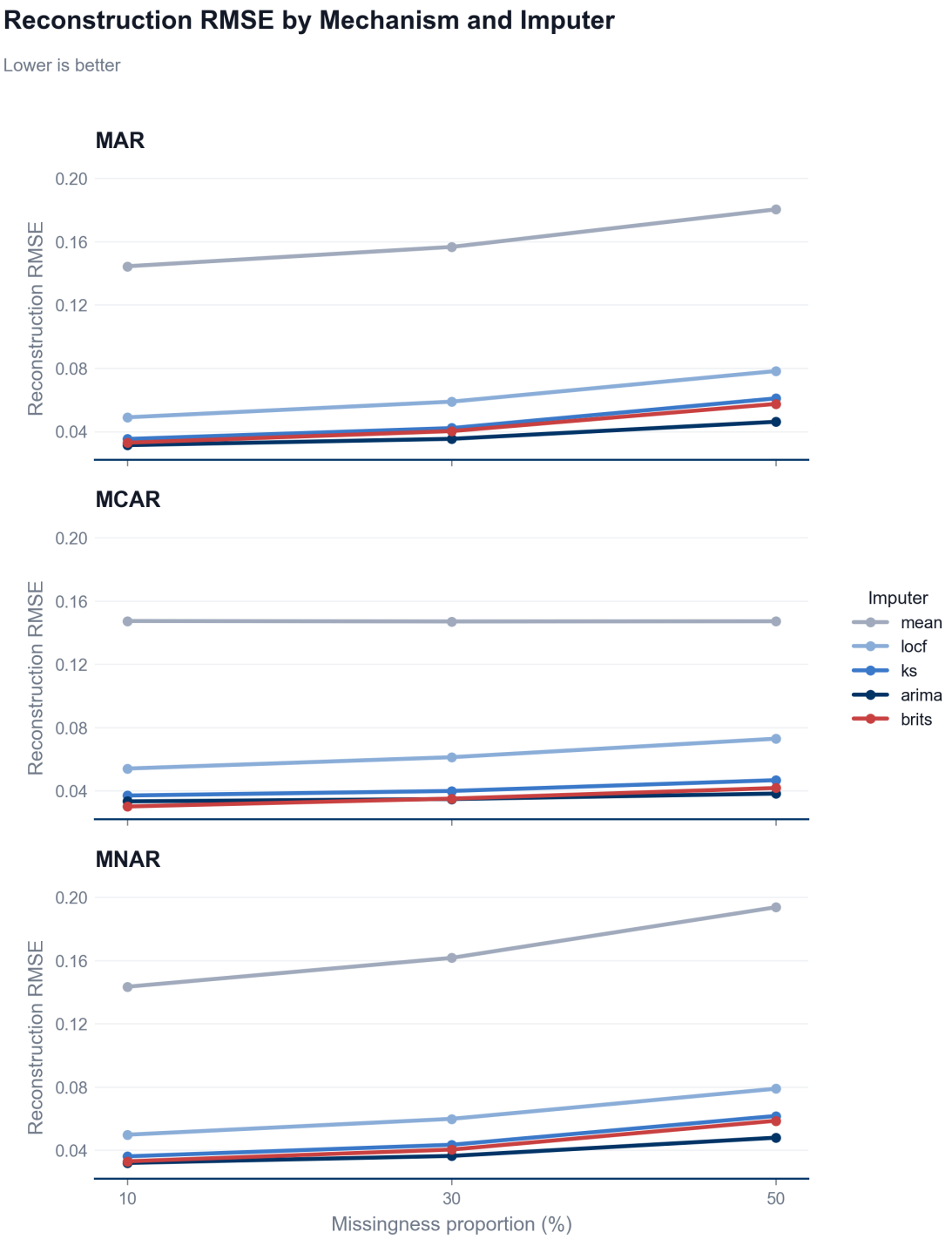


Figure D.13. Comparison of RMSE reconstruction by mechanisms and imputers, for varying missingness proportions.

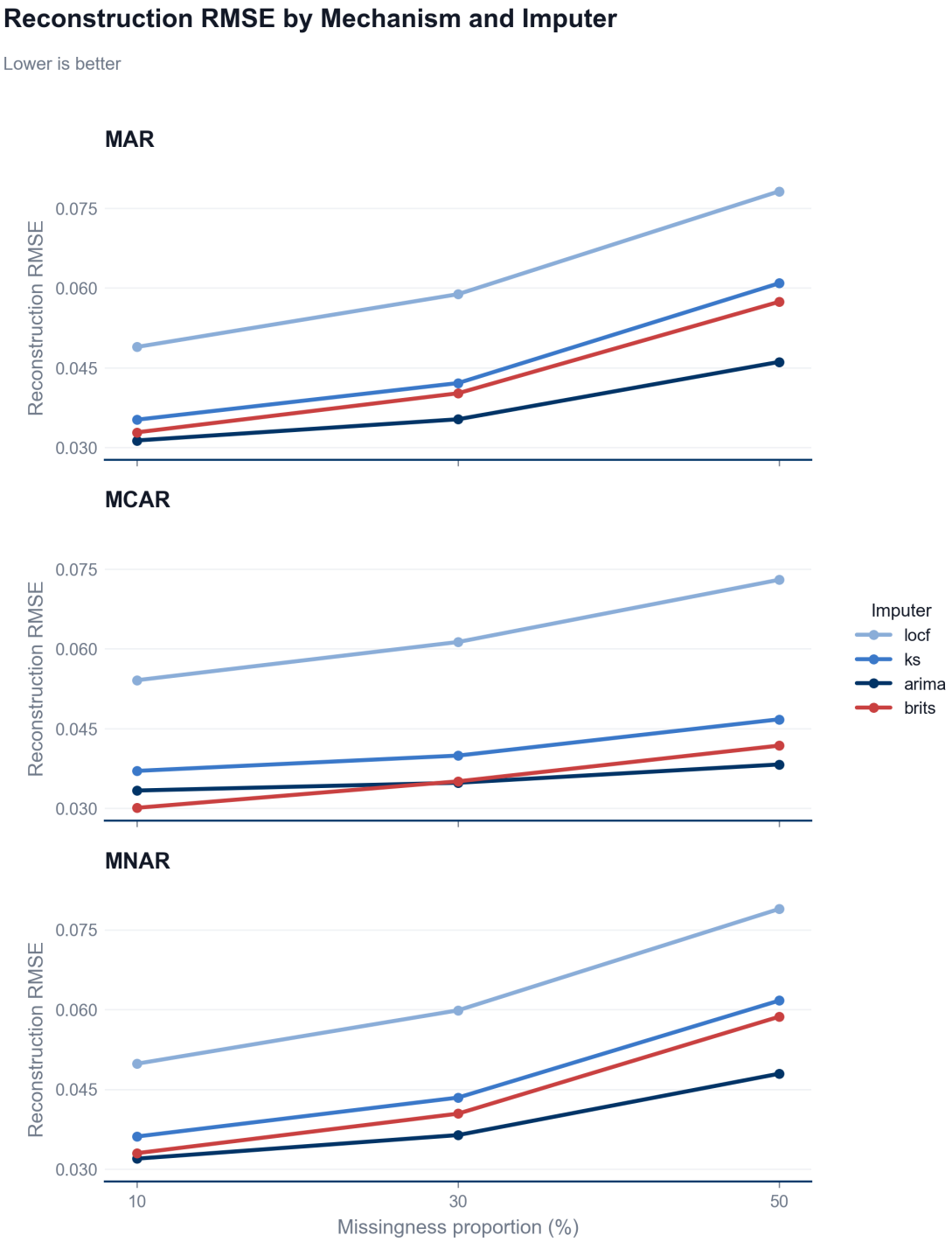


Figure D.14. Comparison of RMSE reconstruction by mechanisms and imputers, excluding mean, for varying missingness proportions.

Tables

Table D.1. Performance metrics (RMSE and MAE) across different imputers, missingness mechanisms, and proportions.

Imputer	Mechanism	Proportion	RMSE	MAE
ARIMA	MAR	10	0.031333	0.020876
		30	0.035314	0.023768
		50	0.046100	0.031050
	MCAR	10	0.033346	0.021317
		30	0.034806	0.022723
		50	0.038249	0.025275
	MNAR	10	0.031967	0.021326
		30	0.036399	0.024320
		50	0.047935	0.031628
BRITS	MAR	10	0.032858	0.018767
		30	0.040205	0.024786
		50	0.057405	0.036837
	MCAR	10	0.030094	0.017397
		30	0.035074	0.021336
		50	0.041793	0.026626
	MNAR	10	0.032994	0.019129
		30	0.040426	0.024888
		50	0.058653	0.037002
KS	MAR	10	0.035238	0.023531
		30	0.042119	0.028243
		50	0.060891	0.040751
	MCAR	10	0.037036	0.024020
		30	0.039914	0.026227
		50	0.046734	0.030576
	MNAR	10	0.036130	0.024240
		30	0.043445	0.029063
		50	0.061681	0.041215
LOCF	MAR	10	0.048911	0.032286
		30	0.058830	0.038996
		50	0.078174	0.053210
	MCAR	10	0.054085	0.034353
		30	0.061268	0.038359
		50	0.072965	0.045074
	MNAR	10	0.049814	0.033123
		30	0.059849	0.039895
		50	0.078907	0.053790
Mean	MAR	10	0.144442	0.112215
		30	0.156675	0.125496
		50	0.180419	0.150742
	MCAR	10	0.147382	0.110098
		30	0.147117	0.109911
		50	0.147228	0.110014
	MNAR	10	0.143391	0.110934
		30	0.161636	0.131070
		50	0.193669	0.167215

Table D.2. Bias and variance metrics for the XGB forecaster across different imputers, missingness mechanisms, and proportions.

Imputer	Mechanism	Proportion	Bias vs Truth	Bias vs Baseline	Variance
ARIMA	MAR	10	83.406760	11.492931	9.363059
		30	42.178408	1.232544	5.019903
		50	28.881647	2.642239	4.133067
	MCAR	10	58.628552	4.637808	12.804341
		30	29.626694	3.393108	2.821089
		50	48.833055	3.209227	2.907620
	MNAR	10	56.188678	4.122224	4.321586
		30	44.945024	1.963653	9.718227
		50	36.174503	1.217215	3.604195
BRITS	MAR	10	70.984081	6.721563	23.694686
		30	90.269949	13.734485	30.836958
		50	70.868277	8.193824	10.016963
	MCAR	10	61.723796	5.349082	4.186864
		30	46.150397	2.148698	8.643686
		50	49.782979	4.360251	5.095404
	MNAR	10	65.015764	5.562239	4.569156
		30	46.565446	2.341166	12.843997
		50	34.095878	1.727892	9.050056
KS	MAR	10	53.774945	3.010762	10.141844
		30	34.027406	2.832137	2.071191
		50	30.562077	2.172714	4.531780
	MCAR	10	38.746136	2.224068	6.539311
		30	58.463608	4.088803	9.815597
		50	22.321448	5.001800	3.573291
	MNAR	10	63.710528	5.923129	8.689409
		30	29.464396	3.854999	14.663873
		50	25.789071	3.405304	6.802068
LOCF	MAR	10	55.084742	2.988425	10.307938
		30	58.956848	4.295771	9.801049
		50	54.069345	4.214719	7.215326
	MCAR	10	53.186621	3.089084	9.138726
		30	85.349248	11.848321	7.552531
		50	85.313691	11.082907	10.201562
	MNAR	10	56.188188	3.794701	6.522550
		30	66.288084	5.960724	14.615307
		50	42.452891	1.996680	3.776329
Mean	MAR	10	236.425845	91.449791	11.075640
		30	209.509874	74.807182	3.414779
		50	224.612718	83.719429	36.998803
	MCAR	10	183.861770	57.628423	14.332386
		30	202.851599	71.712400	6.863648
		50	228.788288	87.023626	6.899525
	MNAR	10	109.073525	23.090607	28.383159
		30	147.589488	40.275772	13.160097
		50	96.637256	18.006606	16.692698

Table D.3. Bias and variance metrics for the ES forecaster across different imputers, missingness mechanisms, and proportions.

Imputer	Mechanism	Proportion	Bias vs Truth	Bias vs Baseline	Variance
ARIMA	MAR	10	26.929200	0.002515	0.005897
		30	25.655398	0.030896	0.036164
		50	25.072277	0.063060	0.053265
	MCAR	10	27.321411	0.000382	0.005156
		30	27.239978	0.000981	0.008075
		50	28.922228	0.024204	0.043774
	MNAR	10	27.384280	0.000328	0.000379
		30	27.311074	0.001285	0.003558
		50	26.730459	0.013265	0.017845
BRITS	MAR	10	28.935509	0.037466	0.019273
		30	52.004931	4.047348	4.640838
		50	54.563303	4.884903	3.318221
	MCAR	10	28.813132	0.027182	0.018532
		30	32.715753	0.358323	0.141972
		50	31.616479	0.380715	2.025304
	MNAR	10	28.228002	0.011070	0.002968
		30	32.035629	0.279274	0.012086
		50	30.512238	0.151633	0.016180
KS	MAR	10	26.879033	0.004079	0.003881
		30	27.634426	0.009708	0.025922
		50	29.031613	0.057531	0.017943
	MCAR	10	27.185306	0.001033	0.003184
		30	28.985271	0.058240	0.036834
		50	32.700098	0.287249	0.011090
	MNAR	10	27.198365	0.000741	0.000362
		30	28.752889	0.042084	0.037609
		50	28.593149	0.042737	0.000758
LOCF	MAR	10	26.951998	0.005334	0.004571
		30	28.278702	0.048247	0.031065
		50	26.128744	0.076553	0.019063
	MCAR	10	27.251953	0.002498	0.004812
		30	28.980228	0.072405	0.024216
		50	28.401789	0.076957	0.004248
	MNAR	10	27.251717	0.001545	0.000864
		30	29.516918	0.098221	0.001053
		50	26.651523	0.064951	0.002117
Mean	MAR	10	46.846435	2.642289	3.084621
		30	99.627201	22.951494	2.072037
		50	135.592806	42.082753	0.376151
	MCAR	10	33.924005	0.397539	0.661322
		30	55.999480	5.168300	1.219578
		50	93.295805	19.922461	0.796796
	MNAR	10	24.059894	0.187777	0.013583
		30	21.963844	0.333913	0.058984
		50	21.312067	0.473521	0.038072

APPENDIX E

Code Listings

```
def _create_mar_mask(df:pd.DataFrame, n:float) -> pd.DataFrame:
    """
    Creates a missingness mask using synthetic MAR. Probability
       of missingness uses most
    correlated features.
    """
    N = int((n / 100) * df.size)
    arr = df.to_numpy(dtype=float)

    # Calculate feature Pearson correlations.
    corr = np.abs(np.corrcoef(arr, rowvar=False))

    # Fill diagonal with -inf to prevent self corr.
    np.fill_diagonal(corr, -np.inf)

    # List of indices for most correlated feature for each
       feature.
    drivers = corr.argmax(axis=1)

    # Create the weights matrix - w[i, j] is how much df[i,j]
       wants to be missing.
    W = np.empty_like(arr)

    # Column j, driver d.
    for j, d in enumerate(drivers):
```

```
        # Calculate z scores by driver column.
        zd = (arr[:, d] - arr[:, d].mean()) / arr[:, d].std()

        # Create weights matrix using sigmoid function.
        W[:, j] = 1.0 / (1.0 + np.exp(-K * zd))

    w = W.flatten()

    # Convert weights into probabilities.
    w = w / w.sum()

    # Randomly select without repeats using probability weights.
    idx = np.random.choice(w.size, N, replace=False, p=w)

    mask = np.zeros(w.size, dtype=bool)
    mask[idx] = True
    mask = mask.reshape(df.shape)

    return pd.DataFrame(mask, columns=df.columns,
       index=df.index)
```

Listing E.1. Synthetic MAR mask function.

```
def _create_mcar_mask(df:pd.DataFrame, n:float) -> pd.DataFrame:
    """
    Creates a missingness mask using synthetic MCAR.
    """
    N = int((n / 100) * df.size)
    arr = df.to_numpy(dtype=float)

```

```
    mask = np.zeros(len(arr.flatten()), dtype=bool)

    # Randomly select without repeats.
    idx = np.random.choice(len(df.to_numpy().flatten()), N,
        replace=False)

    mask[idx] = True
    mask = mask.reshape(df.shape)

    return pd.DataFrame(mask, columns=df.columns,
        index=df.index)
```

Listing E.2. Synthetic MCAR mask function.

```
def _create_mnar_mask(df:pd.DataFrame, n:float) -> pd.DataFrame:
    """
    Synthetic MNAR: missingness in each cell is driven by its
        own value,
    through a per-column-standardised logistic.
    """
    N = int((n / 100) * df.size)
    arr = df.to_numpy(dtype=float)

    # Calculate z-score per column.
    Z = (arr - arr.mean(axis=0)) / arr.std(axis=0)
    W = 1 / (1 + np.exp(-K * Z))

    w = W.flatten()

    # Convert weights into probabilities.
    w = w / w.sum()
```

```
17
18      # Randomly select without repeats using probability weights.
19      idx = np.random.choice(w.size, N, replace=False, p=w)
20
21      mask = np.zeros(w.size, dtype=bool)
22      mask[idx] = True
23      mask = mask.reshape(df.shape)
24
25      return pd.DataFrame(mask, columns=df.columns,
            index=df.index)
```

Listing E.3. Synthetic MNAR mask function.